Vellore Institute of Technology, Chennai
BECE407P - ASIC Design

Lab Report: 4-Stage Pipelined CPU

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1. Aim:

To design, verify, and implement the complete RTL-to-GDSII flow for the control logic of a simplified 4-stage pipelined CPU (Fetch, Decode, Execute, Writeback). The design must be in synthesizable Verilog and include stall logic to handle Read-After-Write (RAW) data hazards.

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2. EDA Tools Used:

- **Simulation (Functional Verification):** Cadence Xcelium
- **Code Coverage:** Cadence Nclaunch & Incisive Metric Center (IMC)
- **Synthesis & DFT:** Cadence Genus
- **Logic Equivalence Check (LEC):** Cadence Conformal
- **ATPG:** Cadence Modus
- **Physical Design (Place & Route):** Cadence Innovus

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3. Detailed description of the Designs:

The design implements a 4-stage CPU pipeline: Fetch (IF), Decode (ID), Execute (EX), and Writeback (WB)..

Key Components:

- **Pipeline Registers:** if_id_instr, id_ex_*, and ex_wb_* registers hold data between stages.

- **Program ROM:** To ensure synthesizability, the program memory is implemented as a combinational case statement in the Fetch stage, mapping pc values to specific instructions. This avoids the non-synthesizable initial block.

- **Hazard Detection Unit:** This is the core control logic. It consists of combinational logic that detects two types of RAW hazards:

1. **EX/ID Hazard (hazard_ex):** Detects if the instruction in ID is reading a register that the instruction in EX is writing to.

2. **WB/ID Hazard (hazard_wb):** Detects if the instruction in ID is reading a register that the instruction in WB is writing to. This check prevents a read/write race condition on the register file.

- **Stall Logic:** If either hazard is detected (hazard_detected is high), the unit asserts stall signals:
    - pc_stall: Freezes the Program Counter.
    - if_id_stall: Prevents the IF/ID register from loading the next (stalled) instruction.
    - id_ex_bubble: Injects a NOP into the ID/EX register, allowing the hazardous instruction to complete before the dependent instruction proceeds.
- **Synthesis & LEC Considerations:**
    - An output port debug_reg_r4 was added and assigned to reg_file[4]. This creates a logic cone to an output, preventing Genus from optimizing away the entire pipeline as "unused."
    - All flip-flops in the design, including the pipeline registers and the register file, were implemented with a consistent **asynchronous active-high reset** (posedge rst). This was a critical fix to resolve LEC failures, which occurred when the register file's synchronous reset logic was modeled differently from the pipeline's asynchronous reset.

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4. Procedure:

The full RTL-to-GDSII flow was executed.
Step 4.1: Functional Verification (**Xcelium**)
The RTL was first verified to ensure logical correctness. The testbench (pipeline_control_tb.v) was designed to specifically trigger the RAW hazard on register R1.

- **Command:** xrun pipeline_control.v pipeline_control_tb.v -access +rwc -sv
- **Result:** The simulation passed, with the testbench console printing TEST PASSED: Hazard detected and resolved correctly. This confirms the stall logic worked, allowing the final value of R4 (22) to be calculated correctly.

Step 4.2: Code Coverage (**Nclaunch & Xcelium IMC**)
To ensure the testbench provided adequate stimulus, code coverage was run.

- **Command:** nclaunch

- **Result:** The Incisive Metric Center (IMC) showing coverage for all metrics (Block, Toggle, Expression), confirming that the hazard logic and all pipeline stages were exercised.

Step 4.3: Synthesis & DFT (**Genus**)
The verified RTL was synthesized into a gate-level netlist using the pipeline_control_script_dft.tcl script.

- **Command:** genus -f pipeline_control_script_dft.tcl
- **Process:**
    1. The script read the 90nm slow.lib, the RTL (pipeline_control.v), and the timing constraints (pipeline_control_sdc.g).

    2. DFT (Design For Test) logic was configured, defining a muxed_scan style and a SE (Scan Enable) port.
    3. The design was synthesized using syn_generic, syn_map, and syn_opt.
    4. A single scan chain (top_chain) was defined and connected to all flops using connect_scan_chains.
    5. The final DFT-inserted netlist (pipeline_control_generated_netlist_dft.v) and new SDC file (.sdc) were written out.
    6. ATPG (Automatic Test Pattern Generation) files were generated by write_dft_atpg and saved in the test_scripts/ directory.

Step 4.4: Logic Equivalence Check (**Conformal**)
LEC was used to mathematically prove that the synthesized netlist was logically identical to the original RTL.

- **Command:** lec -Xl -nogui -color -64 -dofile pipeline_control_lec.do
- **Process:** The pipeline_control_lec.do script was configured to read the golden (RTL) and revised (netlist) designs. It was instructed to ignore the DFT pins (scan_in, scan_out) and constrain the SE pin to 0 to keep the netlist in its functional mode.
- **Result:** The comparison passed.

Step 4.5: ATPG (**Modus**)
The ATPG files generated by Genus were used to generate test patterns.

- **Directory:** synthesis/test_scripts/
- **Command:** modus -f pipeline_control_modus.tcl
- **Result:** Modus successfully generated test vectors, confirming the scan chain was correctly inserted and the design is testable.

Step 4.6: Physical Design (**Innovus**)
The final, verified netlist was used for Place & Route.

- **Command:** innovus (GUI/Tcl commands)
- **Process:**
    1. **Import:** The **pipeline_control_generated_netlist_dft.v** and

pipeline_control_generated_sdc_dft.sdc files were imported. MMMC was configured using **slow.lib, fast.lib**, and the **GPDK 90nm LEF files**.

2. **Floorplan:** The design was floorplanned with a 0.7 core utilization and an aspect ratio of 1.
3. **Power:** Power rings (Metal8/9) and stripes (Metal8) were added for vdd and vss.
4. **Placement:** Standard cells were placed using place_opt_design.
5. **CTS:** The clock tree was synthesized using ccopt_design.
6. **Routing:** route_design (NanoRoute) was run to route the design.
7. **Post-Route:** Filler cells were added to complete the layout.
8. **Export:** The final layout was exported as **pipeline_control.gds.**

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5. RTL Codes:

- **Pipeline_control.v**

```
`timescale 1ns / 1ps

// 4-Stage Pipelined CPU (IF, ID, EX, WB) with Hazard Detection
// Simplified Instruction Set:
// ADD Rd, Rs1, Rs2 -> Opcode: 2'b00
// NOP              -> Opcode: 2'b11

module pipeline_cpu #(
   // --- Parameters ---
   parameter REG_ADDR_WIDTH = 4,
   parameter INSTR_WIDTH = 16,
   parameter REG_FILE_DEPTH = 16,
   parameter DATA_WIDTH = 32,
   parameter INSTR_MEM_DEPTH = 16
) (
   input wire clk,
   input wire rst,
   output wire [DATA_WIDTH-1:0] debug_reg_r4
);

   // Opcodes
   parameter OP_ADD = 2'b00;
   parameter OP_NOP = 2'b11;

   // --- Register File ---
   reg [DATA_WIDTH-1:0] reg_file [0:REG_FILE_DEPTH-1];

   // --- Program Counter (PC) ---
   reg [REG_ADDR_WIDTH-1:0] pc;

   // --- Pipeline Registers ---
```

```
// IF/ID
reg [INSTR_WIDTH-1:0] if_id_instr;

// ID/EX
reg id_ex_reg_write;
reg [REG_ADDR_WIDTH-1:0] id_ex_rd_addr;
reg [DATA_WIDTH-1:0] id_ex_data1;
reg [DATA_WIDTH-1:0] id_ex_data2;

// EX/WB
reg ex_wb_reg_write;
reg [REG_ADDR_WIDTH-1:0] ex_wb_rd_addr;
reg [DATA_WIDTH-1:0] ex_wb_alu_result;

// --- Hazard Detection Wires ---
wire hazard_detected;
wire pc_stall;
wire if_id_stall;
wire id_ex_bubble;
wire hazard_ex;
wire hazard_wb;

// --- ID Stage Wires (Decode) ---
wire [REG_ADDR_WIDTH-1:0] id_rs1_addr;
wire [REG_ADDR_WIDTH-1:0] id_rs2_addr;
wire [REG_ADDR_WIDTH-1:0] id_rd_addr;
wire id_reg_write;
wire [1:0] id_opcode;

// --- EX Stage Wires (Execute) ---
wire [DATA_WIDTH-1:0] ex_alu_result;

// --- VERILOG-2001 COMPATIBILITY ---
// Declare loop variable here, not in the for-loop
integer i;

// ============================================================
// HAZARD DETECTION UNIT (Combinational)
// ============================================================

// Inputs from ID stage (current instruction being decoded)
assign id_rs1_addr = if_id_instr[9:6];
assign id_rs2_addr = if_id_instr[5:2];

// Hazard between ID and EX stage
assign hazard_ex = id_ex_reg_write && (id_ex_rd_addr != 0) &&
            ((id_ex_rd_addr == id_rs1_addr) || (id_ex_rd_addr == id_rs2_addr));
```

```
// Hazard between ID and WB stage (prevents read/write race)
assign hazard_wb = ex_wb_reg_write && (ex_wb_rd_addr != 0) &&
                ((ex_wb_rd_addr == id_rs1_addr) || (ex_wb_rd_addr == id_rs2_addr));

// Stall if either hazard is detected
assign hazard_detected = hazard_ex || hazard_wb;

// Stall Signals:
assign pc_stall     = hazard_detected; // Stop PC from incrementing
assign if_id_stall  = hazard_detected; // Stop IF/ID reg from loading new instr
assign id_ex_bubble = hazard_detected; // Inject a NOP into the ID/EX register


// ======================================================================
// PIPELINE STAGES (Sequential Logic)
// ======================================================================
always @(posedge clk or posedge rst) begin
   if (rst) begin
      // Reset all pipeline stages
      pc <= 0;
      if_id_instr <= {OP_NOP, 14'b0}; // Reset to NOP

      id_ex_reg_write <= 0;
      id_ex_rd_addr <= 0;
      id_ex_data1 <= 0;
      id_ex_data2 <= 0;

      ex_wb_reg_write <= 0;
      ex_wb_rd_addr <= 0;
      ex_wb_alu_result <= 0;
   end else begin

      // --- IF Stage ---
      if (!pc_stall) begin
         pc <= pc + 1;
      end

      // --- IF/ID Register ---
      if (!if_id_stall) begin
         // Synthesizable ROM: Hardcode the program
         case (pc)
            0: if_id_instr <= {OP_ADD, 4'd1, 4'd2, 4'd3, 2'b00}; // ADD R1, R2, R3
            1: if_id_instr <= {OP_ADD, 4'd4, 4'd1, 4'd5, 2'b00}; // ADD R4, R1, R5
            default: if_id_instr <= {OP_NOP, 14'b0}; // NOP
         endcase
      end

      // --- ID/EX Register ---
```

```
        if (id_ex_bubble) begin
            // Inject NOP (all control signals 0)
            id_ex_reg_write <= 0;
            id_ex_rd_addr <= 0;
            id_ex_data1 <= 0;
            id_ex_data2 <= 0;
        end else begin
            // Load decoded values
            id_ex_reg_write <= id_reg_write;
            id_ex_rd_addr <= id_rd_addr;
            id_ex_data1 <= reg_file[id_rs1_addr];
            id_ex_data2 <= reg_file[id_rs2_addr];
        end

        // --- EX/WB Register ---
        ex_wb_reg_write <= id_ex_reg_write;
        ex_wb_rd_addr <= id_ex_rd_addr;
        ex_wb_alu_result <= ex_alu_result;

    end
end

// --- ID Stage (Combinational Decode) ---
assign id_opcode   = if_id_instr[15:14];
assign id_rd_addr  = if_id_instr[13:10];
assign id_reg_write = (id_opcode == OP_ADD); // Only ADD writes to reg file

// --- EX Stage (Combinational ALU) ---
assign ex_alu_result = id_ex_data1 + id_ex_data2; // Only operation is ADD

// --- WB Stage (Write to Reg File) ---
always @(posedge clk or posedge rst) begin
    if (rst) begin
        // Add asynchronous reset for the entire reg_file
        for (i = 0; i < REG_FILE_DEPTH; i = i + 1) begin
            reg_file[i] <= 0;
        end
    end else if (ex_wb_reg_write && (ex_wb_rd_addr != 0)) begin
        // Synchronous write (no longer gated by !rst)
        reg_file[ex_wb_rd_addr] <= ex_wb_alu_result;
    end
end

assign debug_reg_r4 = reg_file[4];

Endmodule
```

- **Pipeline_control_tb.v**

```
`timescale 1ns / 1ps

module pipeline_control_tb;

    // --- Parameters ---
    localparam CLK_PERIOD = 10;

    // --- Signals ---
    reg clk;
    reg rst;

    // Instantiate the DUT (Design Under Test)
    pipeline_cpu dut (
        .clk(clk),
        .rst(rst)
    );

    // --- Clock Generation ---
    initial begin
        clk = 0;
        forever #(CLK_PERIOD / 2) clk = ~clk;
    end

    // --- Test Sequence ---
    initial begin
        // Dump waves
        $dumpfile("wave.vcd");
        $dumpvars(0, dut);

        // 1. Reset the system
        rst = 1;
        @(negedge clk);
        @(negedge clk);
        rst = 0;

        // 2. Monitor execution
        // We need to run for enough cycles to complete the program
        // Cycle 0: PC=0, IF fetches ADD R1, R2, R3
        // Cycle 1: PC=1, IF fetches ADD R4, R1, R5 | ID decodes ADD R1
        // Cycle 2: PC=2, IF fetches NOP            | ID decodes ADD R4 | EX executes ADD R1
        // Cycle 3: PC=2, IF fetches NOP (STALLED)  | ID decodes ADD R4 (STALLED) | EX exec.
NOP (BUBBLE) | WB writes R1
        // Cycle 4: PC=3, IF fetches NOP            | ID decodes ADD R4 | EX executes ADD R4
        // Cycle 5: PC=4, IF fetches NOP            | ID decodes NOP    | EX executes NOP | WB
writes R4
        // Cycle 6: PC=5, IF fetches NOP            | ID decodes NOP    | EX executes NOP
        // Cycle 7: WB writes NOP (nothing)
```

```
        $display("Time\tPC\tIF/ID_Instr\tID/EX_Op\tEX/WB_Op\tHazard");
        repeat (10) begin
            @(posedge clk);
            $display("%0t\t%d\t%h\t\t%h\t\t%h\t\t%b",
                     $time,
                     dut.pc,
                     dut.if_id_instr,
                     dut.id_ex_rd_addr, // Show dest reg in ID/EX
                     dut.ex_wb_rd_addr, // Show dest reg in EX/WB
                     dut.hazard_detected);
        end

        // 3. Check final register values
        // R1 = R2(5) + R3(10) = 15
        // R4 = R1(15) + R5(7) = 22
        $display("\n--- Final Register Values ---");
        $display("Register R1 (addr 1): %d", dut.reg_file[1]);
        $display("Register R4 (addr 4): %d", dut.reg_file[4]);

        if (dut.reg_file[1] == 15 && dut.reg_file[4] == 22) begin
            $display("TEST PASSED: Hazard detected and resolved correctly.");
        end else begin
            $display("TEST FAILED: Final register values are incorrect.");
        end

        $finish;
    end

endmodule
```

- **Pipeline_control_sdc.g**

```
create_clock -name clk -period 10 -waveform {0 5} [get_ports "clk"]
set_clock_transition -rise 0.1 [get_clocks "clk"]
set_clock_transition -fall 0.1 [get_clocks "clk"]
set_clock_uncertainty 0.01 [get_ports "clk"]
set_input_delay -max 1.0 [get_ports "rst"] -clock [get_clocks "clk"]
set_output_delay -max 1.0 [get_ports "debug_reg_r4"] -clock [get_clocks "clk"]
```

- **Pipeline_control_script_dft.tcl**

```
set_db init_lib_search_path /home/install/FOUNDRY/digital/90nm/dig/lib
set_db init_hdl_search_path /home/student/22BEC1204_ASIC_DESIGN/work
read_libs slow.lib
read_hdl pipeline_control.v
elaborate
read_sdc pipeline_control_sdc.g
```

```
set_db dft_scan_style muxed_scan

set_db dft_prefix dft_

define_shift_enable -name SE -active high -create_port SE

check_dft_rules

set_db syn_generic_effort medium
set_db syn_map_effort medium
set_db syn_opt_effort medium
syn_generic
syn_map
syn_opt

check_dft_rules

set_db design:pipeline_cpu .dft_min_number_of_scan_chains 1

define_scan_chain -name top_chain -sdi scan_in -sdo scan_out -create_ports

connect_scan_chains -auto_create_chains

syn_opt -incr

report_scan_chains

write_dft_atpg -library /home/install/FOUNDRY/digital/90nm/dig/lib

write_hdl > pipeline_control_generated_netlist_dft.v
write_sdf -timescale ns -nonegchecks -recrem split -edges check_edge -setuphold split >
pipeline_control_generated_delays_dft.sdf
write_sdc > pipeline_control_generated_sdc_dft.sdc
write_scandef > pipeline_control_generated_scanDEF.scandef

gui_show
```

- **Pipeline_control_lec.do**

```
set log file pipeline_control_generated_LEC.log -replace
read library ./slowlib_verilog_generated.v -verilog -both
read design /home/student/22BEC1204_ASIC_DESIGN/work/pipeline_control.v -verilog -golden
read design
/home/student/22BEC1204_ASIC_DESIGN/synthesis/pipeline_control_generated_netlist_dft.v
-verilog -revised
add ignored inputs scan_in -revised
add ignored outputs scan_out -revised
add pin constraints 0 SE -revised
```

set system mode lec
add compare point -all
compare

report verification
set gui on

- **Pipeline_control_modus.tcl**

build_model -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -designsource pipeline_cpu.test_netlist.v -techlib /home/student/22BEC1204_ASIC_DESIGN/LEC/slowlib_verilog_generated.v -designtop pipeline_cpu
build_testmode -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN -assignfile pipeline_cpu.FULLSCAN.pinassign
verify_test_structures -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN
report_test_structures -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN
build_faultmodel -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -fullfault yes
create_scanchain_tests -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN -experiment scan
create_logic_tests -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN -experiment logic -effort high
write_vectors -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN -inexperiment logic -language verilog -scanformat serial -outputfilename pipeline_control_modus_generated_results

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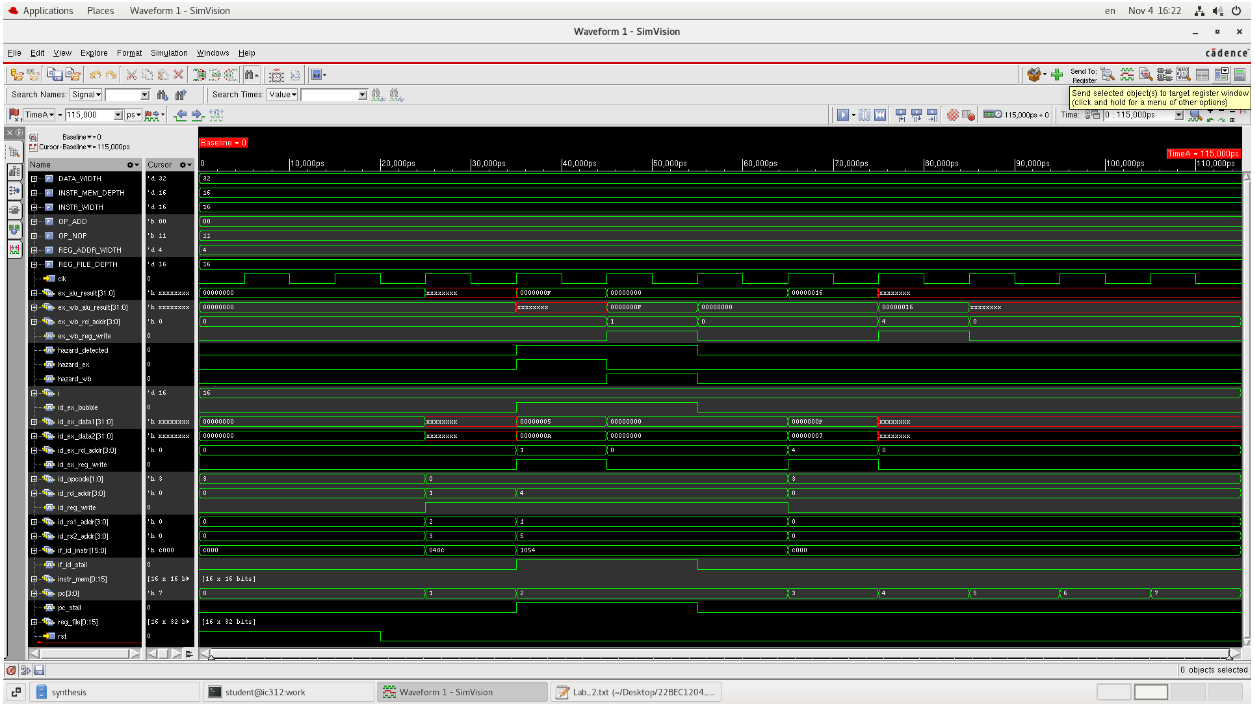
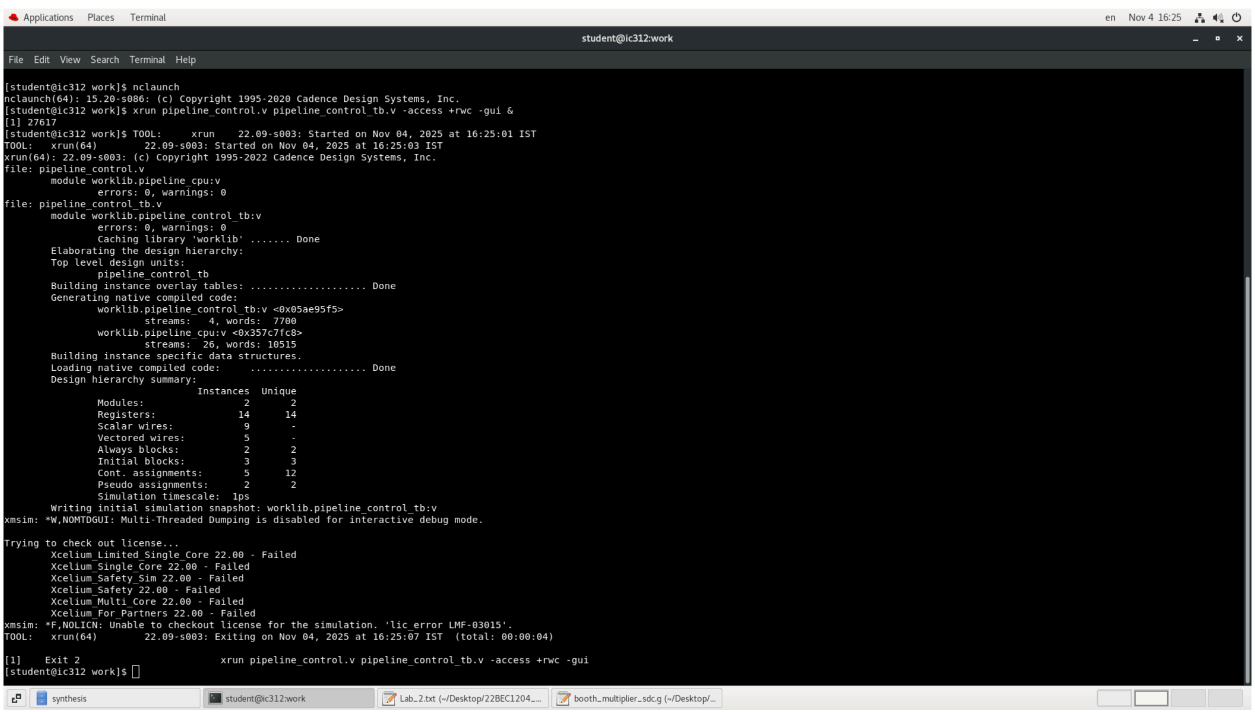
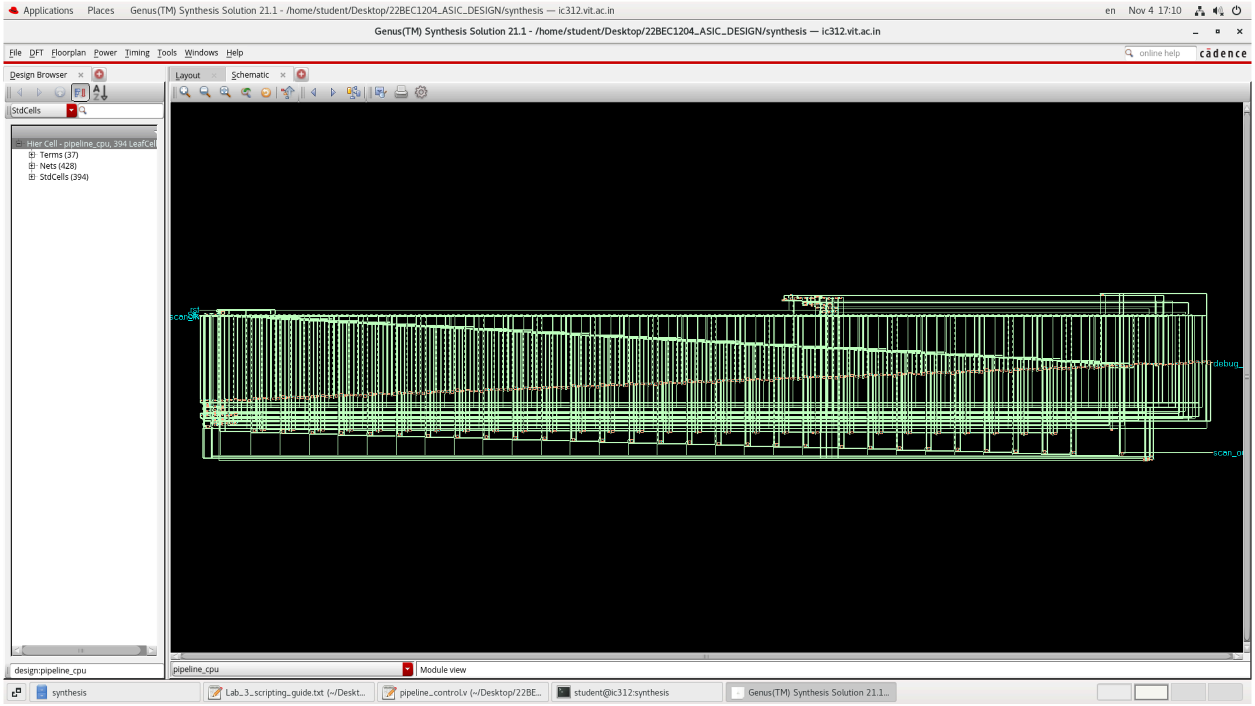
6. Observations (Waveforms & Screenshots):

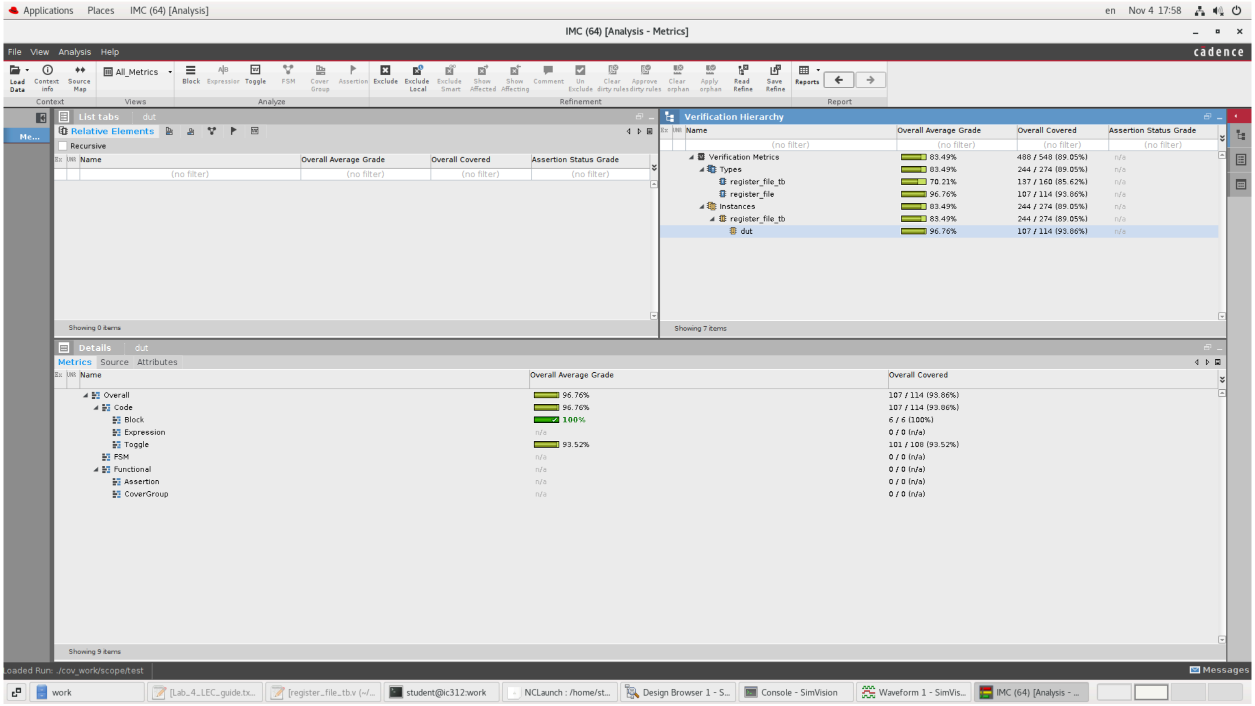


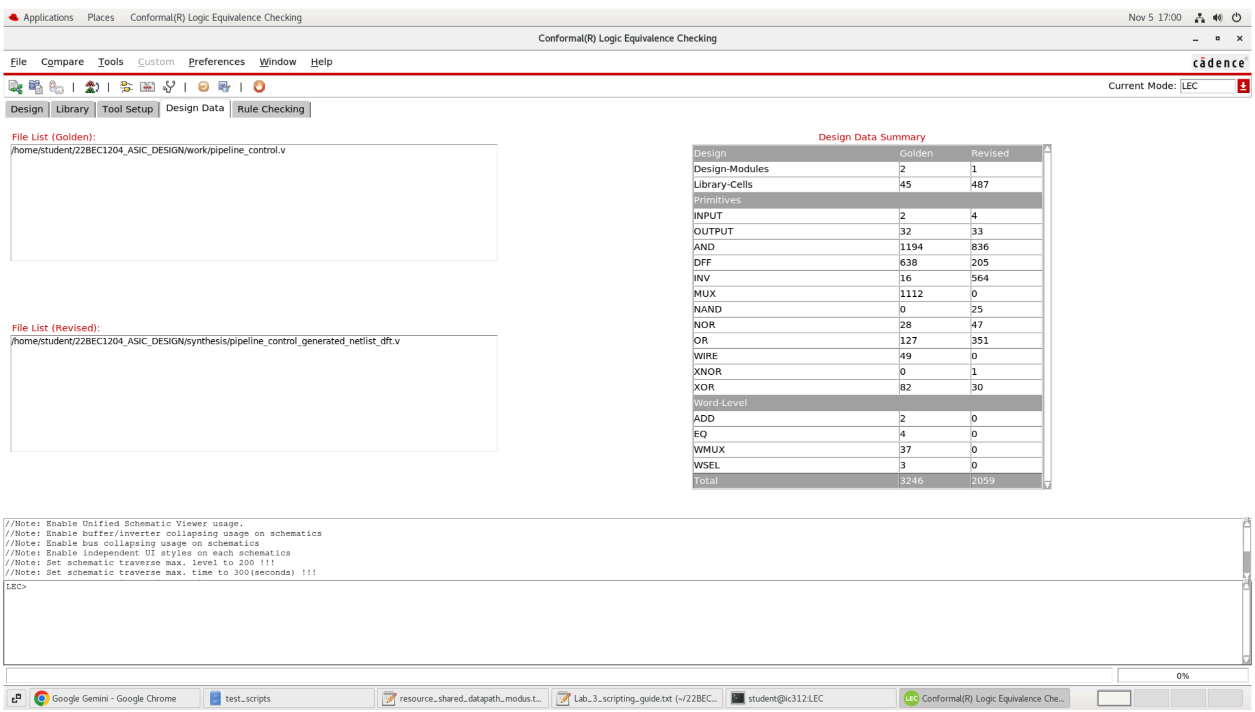


```
[student@ic312 work]$ nclaunch
nclaunch(64): 15.20-s086: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
[student@ic312 work]$ xrun pipeline_control.v pipeline_control_tb.v -access +rwc -gui &
[1] 27617
[student@ic312 work]$ TOOL:     xrun    22.09-s003: Started on Nov 04, 2025 at 16:25:01 IST
TOOL:   xrun(64)        22.09-s003: Started on Nov 04, 2025 at 16:25:03 IST
xrun(64): 22.09-s003: (c) Copyright 1995-2022 Cadence Design Systems, Inc.
file: pipeline_control.v
        module worklib.pipeline_cpu:v
                errors: 0, warnings: 0
file: pipeline_control_tb.v
        module worklib.pipeline_control_tb:v
                errors: 0, warnings: 0
                Caching library 'worklib' ....... Done
        Elaborating the design hierarchy:
        Top level design units:
                pipeline_control_tb
        Building instance overlay tables: .................... Done
        Generating native compiled code:
                worklib.pipeline_control_tb:v <0x05ae95f5>
                        streams:   4, words:  7700
                worklib.pipeline_cpu:v <0x357c7fc8>
                        streams:  26, words: 10515
        Building instance specific data structures.
        Loading native compiled code:     .................... Done
        Design hierarchy summary:
                                 Instances  Unique
                Modules:                 2       2
                Registers:              14      14
                Scalar wires:            9       -
                Vectored wires:          5       -
                Always blocks:           2       2
                Initial blocks:          3       3
                Cont. assignments:       5      12
                Pseudo assignments:      2       2
                Simulation timescale:  1ps
        Writing initial simulation snapshot: worklib.pipeline_control_tb:v
xmsim: *W,NOMTDGUI: Multi-Threaded Dumping is disabled for interactive debug mode.

Trying to check out license...
        Xcelium_Limited_Single_Core 22.00 - Failed
        Xcelium_Single_Core 22.00 - Failed
        Xcelium_Safety_Sim 22.00 - Failed
        Xcelium_Safety 22.00 - Failed
        Xcelium_Multi_Core 22.00 - Failed
        Xcelium_For_Partners 22.00 - Failed
xmsim: *F,NOLICN: Unable to checkout license for the simulation. 'lic_error LMF-03015'.
TOOL:   xrun(64)        22.09-s003: Exiting on Nov 04, 2025 at 16:25:07 IST  (total: 00:00:04)

[1]    Exit 2                     xrun pipeline_control.v pipeline_control_tb.v -access +rwc -gui
[student@ic312 work]$
```




```
================================================================================
                            Verification Report
--------------------------------------------------------------------------------
Category                                                                  Count
--------------------------------------------------------------------------------
1. Non-standard modeling options used:                                      0
--------------------------------------------------------------------------------
2. Incomplete verification:                                                 1
     Command "add ignore outputs" used:                        yes
--------------------------------------------------------------------------------
3. User modification to design:                                             0
--------------------------------------------------------------------------------
4. Conformal Constraint Designer clock domain crossing checks recommended:  0
--------------------------------------------------------------------------------
5. Design ambiguity:                                                        0
--------------------------------------------------------------------------------
6. Compare Results:                                                      PASS
================================================================================
// Command: set gui on
// Switching to GUI mode
[student@ic312 LEC]$
```


IMC (64) [Analysis - Metrics]
File View Analysis Help
List tabs dut
Relative Elements
Recursive
Name
Overall Average Grade
Overall Covered
Assertion Status Grade
Showing 0 items
Verification Hierarchy
Verification Metrics 83.49% 488 / 548 (89.05%) n/a
Types 83.49% 244 / 274 (89.05%) n/a
register_file_tb 70.21% 137 / 160 (85.62%) n/a
register_file 96.76% 107 / 114 (93.86%) n/a
Instances 83.49% 244 / 274 (89.05%) n/a
register_file_tb 83.49% 244 / 274 (89.05%) n/a
dut 96.76% 107 / 114 (93.86%) n/a
Showing 7 items
Details dut
Metrics Source Attributes
Overall 96.76% 107 / 114 (93.86%)
Code 96.76% 107 / 114 (93.86%)
Block 100% 6 / 6 (100%)
Expression n/a 0 / 0 (n/a)
Toggle 93.52% 101 / 108 (93.52%)
FSM n/a 0 / 0 (n/a)
Functional n/a 0 / 0 (n/a)
Assertion n/a 0 / 0 (n/a)
CoverGroup n/a 0 / 0 (n/a)
Showing 9 items
Loaded Run: ./cov_work/scope/test


Conformal(R) Logic Equivalence Checking
File Compare Tools Custom Preferences Window Help
Current Mode: LEC
Design Library Tool Setup Design Data Rule Checking
File List (Golden):
/home/student/22BEC1204_ASIC_DESIGN/work/pipeline_control.v
File List (Revised):
/home/student/22BEC1204_ASIC_DESIGN/synthesis/pipeline_control_generated_netlist_dft.v
Design Data Summary
Design Golden Revised
Design-Modules 2 1
Library-Cells 45 487
Primitives
INPUT 2 4
OUTPUT 32 33
AND 1194 836
DFF 638 205
INV 16 564
MUX 1112 0
NAND 0 25
NOR 28 47
OR 127 351
WIRE 49 0
XNOR 0 1
XOR 82 30
Word-Level
ADD 2 0
EQ 4 0
WMUX 37 0
WSEL 3 0
Total 3246 2059
//Note: Enable Unified Schematic Viewer usage.
//Note: Enable buffer/inverter collapsing usage on schematics
//Note: Enable bus collapsing usage on schematics
//Note: Enable independent UI styles on each schematics
//Note: Set schematic traverse max. level to 200 !!!
//Note: Set schematic traverse max. time to 300(seconds) !!!
LEC>

Design Data Summary

Design	Golden	Revised
Design-Modules	2	1
Library-Cells	45	487
Primitives		
INPUT	2	4
OUTPUT	32	33
AND	1194	836
DFF	638	205
INV	16	564
MUX	1112	0
NAND	0	25
NOR	28	47
OR	127	351
WIRE	49	0
XNOR	0	1
XOR	82	30
Word-Level		
ADD	2	0
EQ	4	0
WMUX	37	0
WSEL	3	0
Total	3246	2059

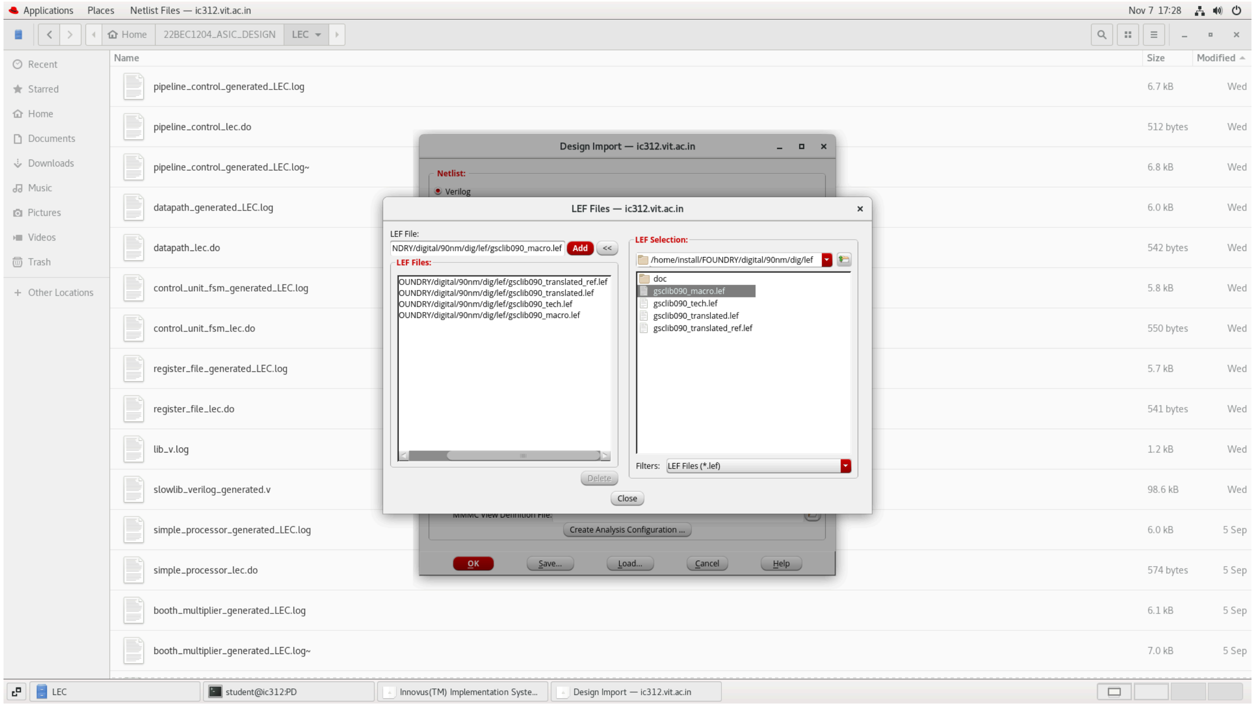
pipeline_control_modus_generated_results.cyclemap

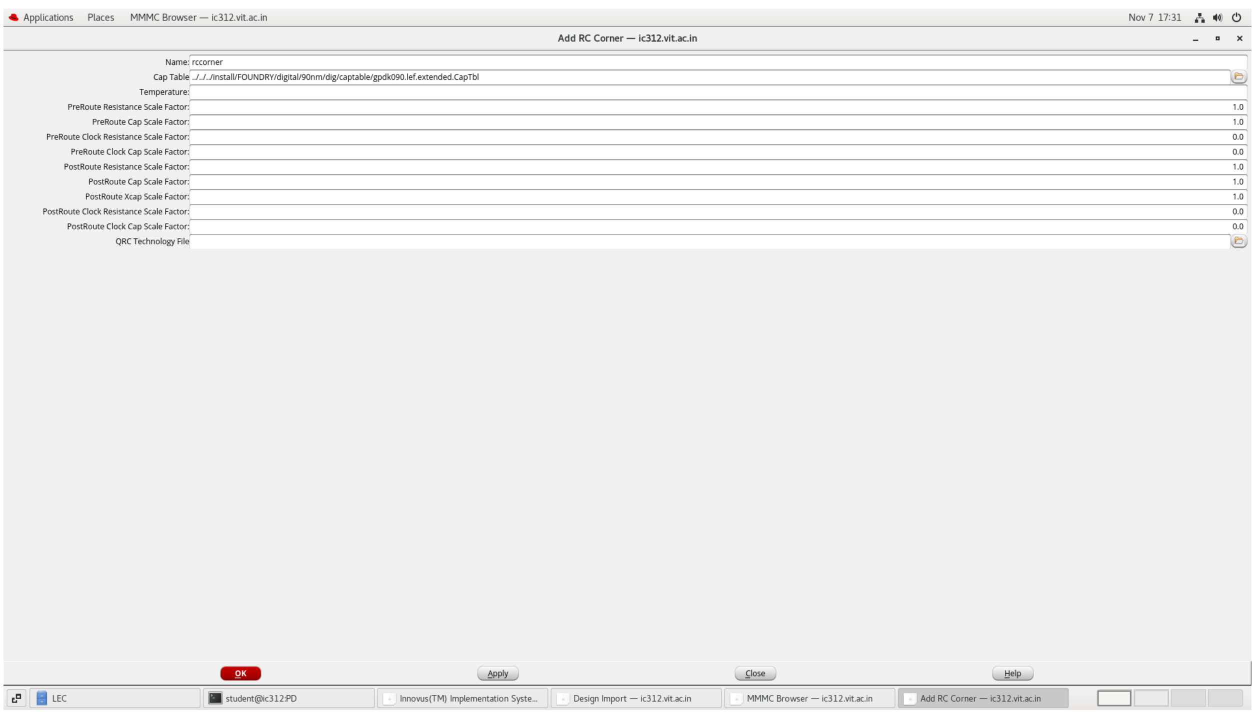
```
//---------------------------------------------------------------------------------------------------------------------------
// Cadence(R) Modus(TM) DFT Software Solution, Version 20.12-s002_1, built Feb 26 2021 (linux26_64)
// This file was created on.............November 05, 2025 at 17:27:49
// The date time stamp for the TBDbin file TBDbin.FULLSCAN.logic;  20251105115748
//-------------------------------------------------------------------------------------------------------------------------------
//-------------------------------------------------------------------------------------------------------------------------------
//-------------------------------------------------------------------------------------------------------------------------------
//   Total    Relative   Relative     Test      CPP       Pattern                 Event                                     Cycle   Scan     Relative   Measure
//   Cycle    Cycle      Simulation   Sequence  Test      Odometer                Type                                      Offset  Length   Scan       Event
//   Count    Count      Time (ns)    Number    Number                                                                                       Number
//-------------------------------------------------------------------------------------------------------------------------------
// The vector file output name:  pipeline_control_modus_generated_results.1.verilog
        3         3         160         1         2    1.1.1.2.1.1            Scan_Load - Shift                              0     205         1        N
      207       207       16560         1         2    1.1.1.2.1.1            end_of_Scan_Load                             204       0         1        N
      209       209       16640         1         2    1.1.1.2.1.2            Scan_Unload - Shift                            0     205         2        N
      413       413       33040         1         2    1.1.1.2.1.2            end_of_Scan_Unload                           204       0         2        Y
// The vector file output name:  pipeline_control_modus_generated_results.2.verilog
      417         3         160         2         4    1.2.1.2.1.1            Scan_Load - Shift                              0     205         1        N
      621       207       16560         2         4    1.2.1.2.1.1            end_of_Scan_Load                             204       0         1        N
      624       210       16720         2         4    1.2.1.2.1.4            Scan_Unload - Shift - Overlap                  0     205         2        Y
      624       210       16720         3         5    1.2.1.3.1.1            Scan_Load - Shift - Overlap                    0     205         2        N
      828       414       33120         3         5    1.2.1.3.1.1            end_of_Scan_Load - Overlap                   204       0         2        Y
      829       415       33120         3         5    1.2.1.3.1.4            Measure_PO                                     0       1         2        Y
      831       417       33280         3         5    1.2.1.3.1.5            Scan_Unload - Shift - Overlap                  0     205         3        Y
      831       417       33280         4         6    1.2.1.3.2.1            Scan_Load - Shift - Overlap                    0     205         3        N
     1035       621       49680         4         6    1.2.1.3.2.1            end_of_Scan_Load - Overlap                   204       0         3        Y
     1036       622       49680         4         6    1.2.1.3.2.4            Measure_PO                                     0       1         3        Y
     1038       624       49840         4         6    1.2.1.3.2.5            Scan_Unload - Shift - Overlap                  0     205         4        Y
     1038       624       49840         5         7    1.2.1.3.3.1            Scan_Load - Shift - Overlap                    0     205         4        N
     1242       828       66240         5         7    1.2.1.3.3.1            end_of_Scan_Load - Overlap                   204       0         4        Y
     1243       829       66240         5         7    1.2.1.3.3.4            Measure_PO                                     0       1         4        Y
     1245       831       66400         5         7    1.2.1.3.3.5            Scan_Unload - Shift - Overlap                  0     205         5        Y
     1245       831       66400         6         8    1.2.1.3.4.1            Scan_Load - Shift - Overlap                    0     205         5        N
     1449      1035       82800         6         8    1.2.1.3.4.1            end_of_Scan_Load - Overlap                   204       0         5        Y
     1450      1036       82800         6         8    1.2.1.3.4.4            Measure_PO                                     0       1         5        Y
     1452      1038       82960         6         8    1.2.1.3.4.5            Scan_Unload - Shift - Overlap                  0     205         6        Y
     1452      1038       82960         7         9    1.2.1.3.5.1            Scan_Load - Shift - Overlap                    0     205         6        N
     1656      1242       99360         7         9    1.2.1.3.5.1            end_of_Scan_Load - Overlap                   204       0         6        Y
     1657      1243       99360         7         9    1.2.1.3.5.4            Measure_PO                                     0       1         6        Y
     1659      1245       99520         7         9    1.2.1.3.5.5            Scan_Unload - Shift - Overlap                  0     205         7        Y
     1659      1245       99520         8        10    1.2.1.3.6.1            Scan_Load - Shift - Overlap                    0     205         7        N
     1863      1449      115920         8        10    1.2.1.3.6.1            end_of_Scan_Load - Overlap                   204       0         7        Y
     1864      1450      115920         8        10    1.2.1.3.6.4            Measure_PO                                     0       1         7        Y
     1866      1452      116080         8        10    1.2.1.3.6.5            Scan_Unload - Shift - Overlap                  0     205         8        Y
     1866      1452      116080         9        11    1.2.1.3.7.1            Scan_Load - Shift - Overlap                    0     205         8        N
     2070      1656      132480         9        11    1.2.1.3.7.1            end_of_Scan_Load - Overlap                   204       0         8        Y
     2071      1657      132480         9        11    1.2.1.3.7.4            Measure_PO                                     0       1         8        Y
     2073      1659      132640         9        11    1.2.1.3.7.5            Scan_Unload - Shift - Overlap                  0     205         9        Y
     2073      1659      132640        10        12    1.2.1.3.8.1            Scan_Load - Shift - Overlap                    0     205         9        N
     2277      1863      149040        10        12    1.2.1.3.8.1            end_of_Scan_Load - Overlap                   204       0         9        Y
```

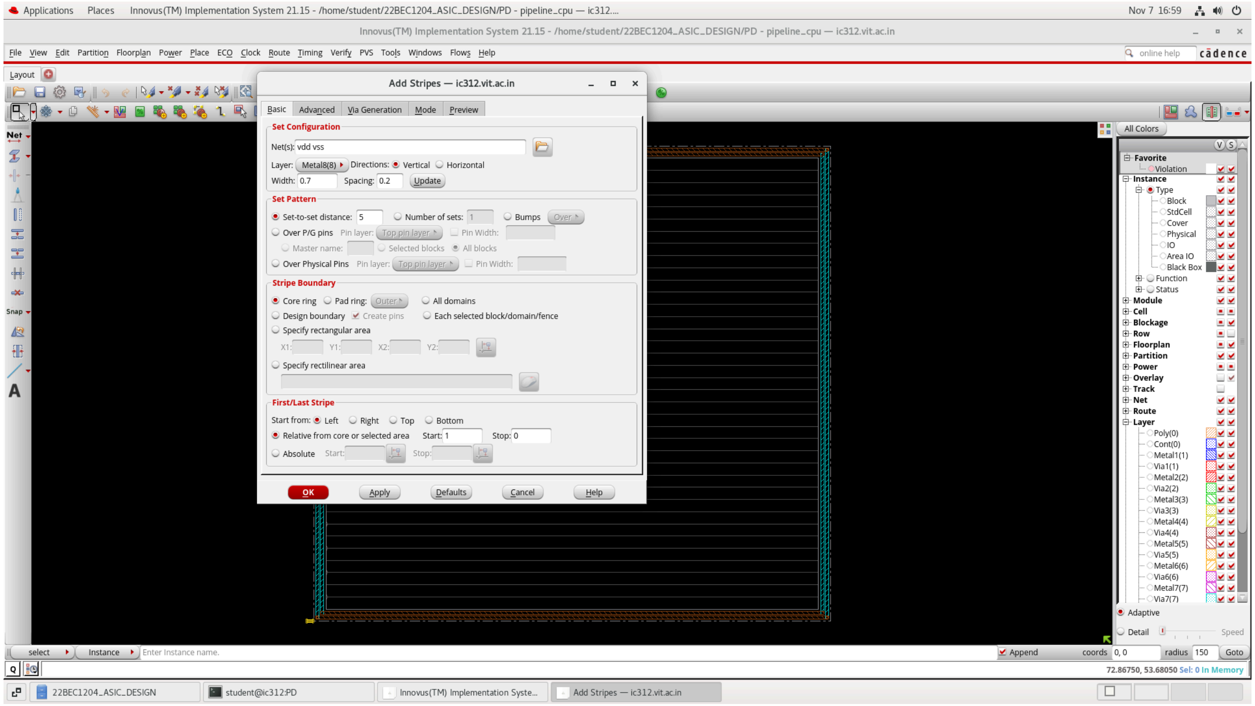
student@ic312:test_scripts

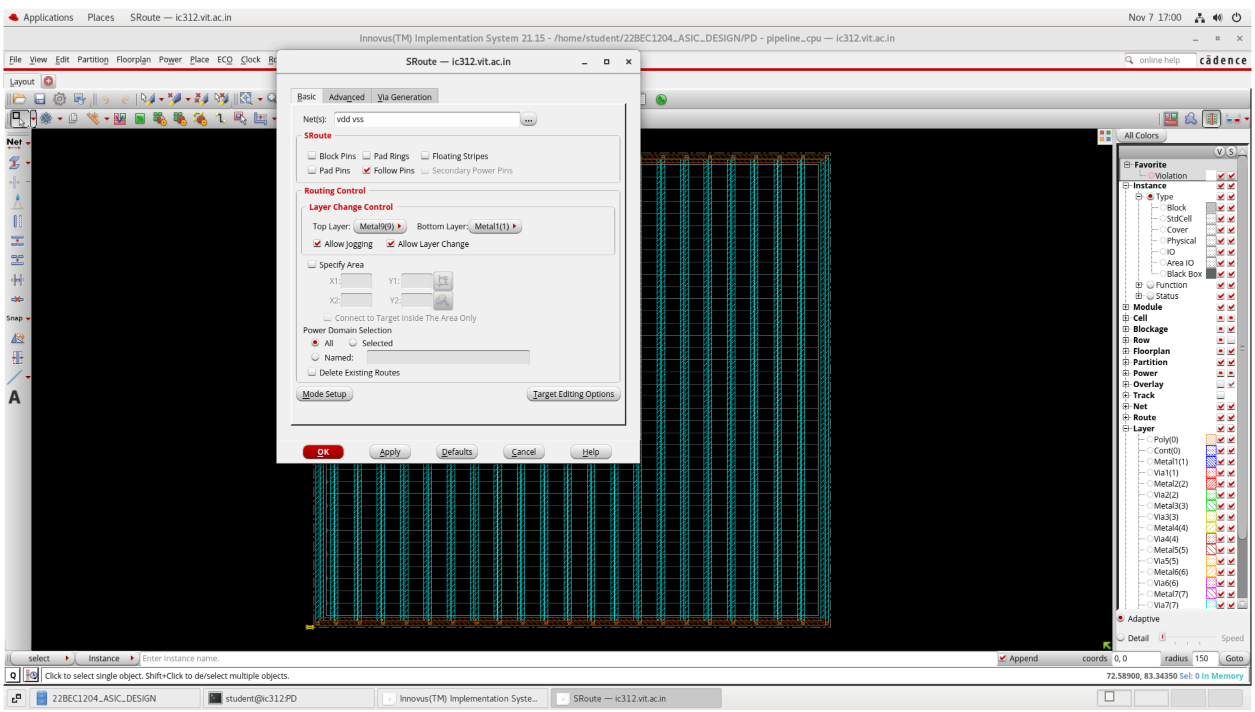
```
     1.2.1.3.9 |  97.16   |  1.13    |  97.16   |  29.00   |  0.00    |  29.00   |        207 |        206 |       2691 |
    1.2.1.3.10 |  97.74   |  0.58    |  97.74   |  29.00   |  0.00    |  29.00   |        207 |        206 |       2898 |
    1.2.1.3.11 |  99.00   |  1.26    |  99.00   |  29.00   |  0.00    |  29.00   |        207 |        206 |       3105 |
    1.2.1.3.12 |  99.25   |  0.25    |  99.25   |  29.00   |  0.00    |  29.00   |        207 |        206 |       3312 |
    1.2.1.3.13 |  99.53   |  0.28    |  99.53   |  29.00   |  0.00    |  29.00   |        207 |        206 |       3519 |
    1.2.1.3.14 |  99.67   |  0.14    |  99.67   |  29.00   |  0.00    |  29.00   |        207 |        206 |       3726 |
    1.2.1.3.15 |  99.77   |  0.10    |  99.77   |  29.00   |  0.00    |  29.00   |        207 |        206 |       3933 |
    1.2.1.3.16 |  99.82   |  0.06    |  99.82   |  29.00   |  0.00    |  29.00   |        207 |        206 |       4140 |
     1.2.1.4.1 |  99.85   |  0.03    |  99.85   |  29.00   |  0.00    |  29.00   |        207 |        206 |       4347 |
     1.2.1.4.2 |  99.86   |  0.01    |  99.86   |  29.00   |  0.00    |  29.00   |        207 |        206 |       4554 |
     1.2.1.4.3 |  99.93   |  0.07    |  99.93   |  29.00   |  0.00    |  29.00   |        207 |        206 |       4761 |
     1.2.1.4.4 |  99.95   |  0.02    |  99.95   |  29.00   |  0.00    |  29.00   |        207 |        206 |       4968 |
     1.2.1.4.5 |  99.96   |  0.01    |  99.96   |  29.00   |  0.00    |  29.00   |        207 |        206 |       5175 |
     1.2.1.4.6 |  99.98   |  0.02    |  99.98   |  29.00   |  0.00    |  29.00   |        207 |        206 |       5382 |
     1.2.1.4.7 |  100.00   |  0.02    |  100.00   |  29.00   |  0.00    |  29.00   |        208 |          0 |       5590 |
  [end TVE_050]

INFO (TDA-001): Maximum Memory used during the run and Cumulative Time in hours:minutes:seconds:

                  Total Memory =            8,345,952  bytes

                      CPU Time =    0:00:00.00
                  Elapsed Time =    0:00:00.26                    [end TDA_001]

     Date Ended:  Wednesday Nov 05 17:27:49 2025  IST

INFO (TVE-002): Verilog write vectors has completed.   [end TVE_002]


-------------------------------------------------------------------------------
*                      Message Summary                                        *
-------------------------------------------------------------------------------
 Count  Number             First Instance of Message Text
------- ------             ------------------------------

  INFO Messages...
      1 INFO (TVE-001): Verilog write vectors started.
      1 INFO (TVE-002): Verilog write vectors has completed.
      4 INFO (TVE-003): Verilog write vectors output file will be: /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts/testresults/verilog/pipeline_control_modus_generated_results.cyclemap.
      2 INFO (TVE-004): Reading test section 1.1.  Test section type equals scan.
      2 INFO (TVE-005): Created 414 total cycles, of which 4 are test cycles, 410 are scan cycles, 0 are dynamic timed cycles and 0 are dynamic cycles that are not timed.
      2 INFO (TVE-008): Created 205 total measures, of which 0 are PO measures and 205 are SO (Scan Out) measures.
      1 INFO (TVE-050):         TEST SEQUENCE COVERAGE ESTIMATE REPORT
      3 INFO (TVE-103): There was no specified TEST offset for this clock 'pin clk'.  A default clock offset of 8.000000 ns will be used.

 For a detailed explanation of a message and a suggested user response execute 'msgHelp <message id>'.  For example: msgHelp TDA-009

-------------------------------------------------------------------------------
#@ End verbose source pipeline_control_modus.tcl
0
@modus:root:/ 2>
```

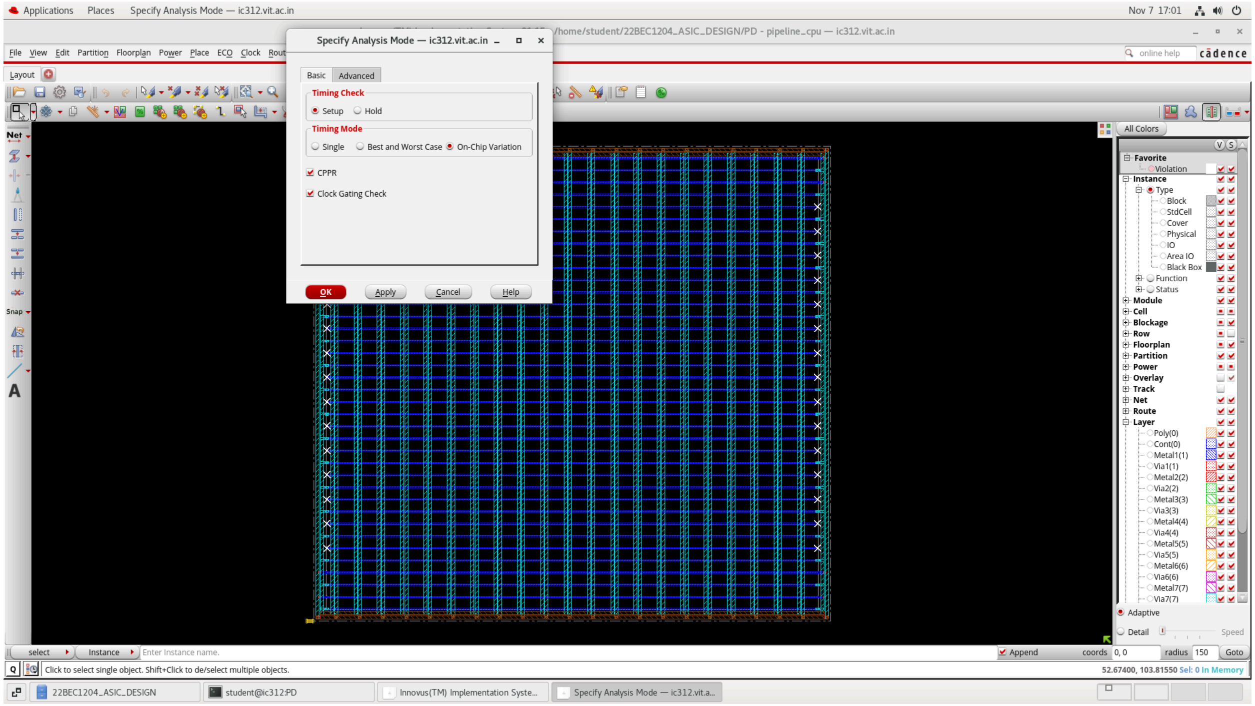

```
-------------------------------------------------------------------------------
*                      Message Summary                                        *
-------------------------------------------------------------------------------
 Count  Number             First Instance of Message Text
------- ------             ------------------------------

  INFO Messages...
      1 INFO (TVE-001): Verilog write vectors started.
      1 INFO (TVE-002): Verilog write vectors has completed.
      4 INFO (TVE-003): Verilog write vectors output file will be: /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts/testresults/verilog/pipeline_control_modus_generated_results.cyclemap.
      2 INFO (TVE-004): Reading test section 1.1.  Test section type equals scan.
      2 INFO (TVE-005): Created 414 total cycles, of which 4 are test cycles, 410 are scan cycles, 0 are dynamic timed cycles and 0 are dynamic cycles that are not timed.
      2 INFO (TVE-008): Created 205 total measures, of which 0 are PO measures and 205 are SO (Scan Out) measures.
      1 INFO (TVE-050):         TEST SEQUENCE COVERAGE ESTIMATE REPORT
      3 INFO (TVE-103): There was no specified TEST offset for this clock 'pin clk'.  A default clock offset of 8.000000 ns will be used.

 For a detailed explanation of a message and a suggested user response execute 'msgHelp <message id>'.  For example: msgHelp TDA-009

-------------------------------------------------------------------------------
#@ End verbose source pipeline_control_modus.tcl
0
@modus:root:/ 2>
```

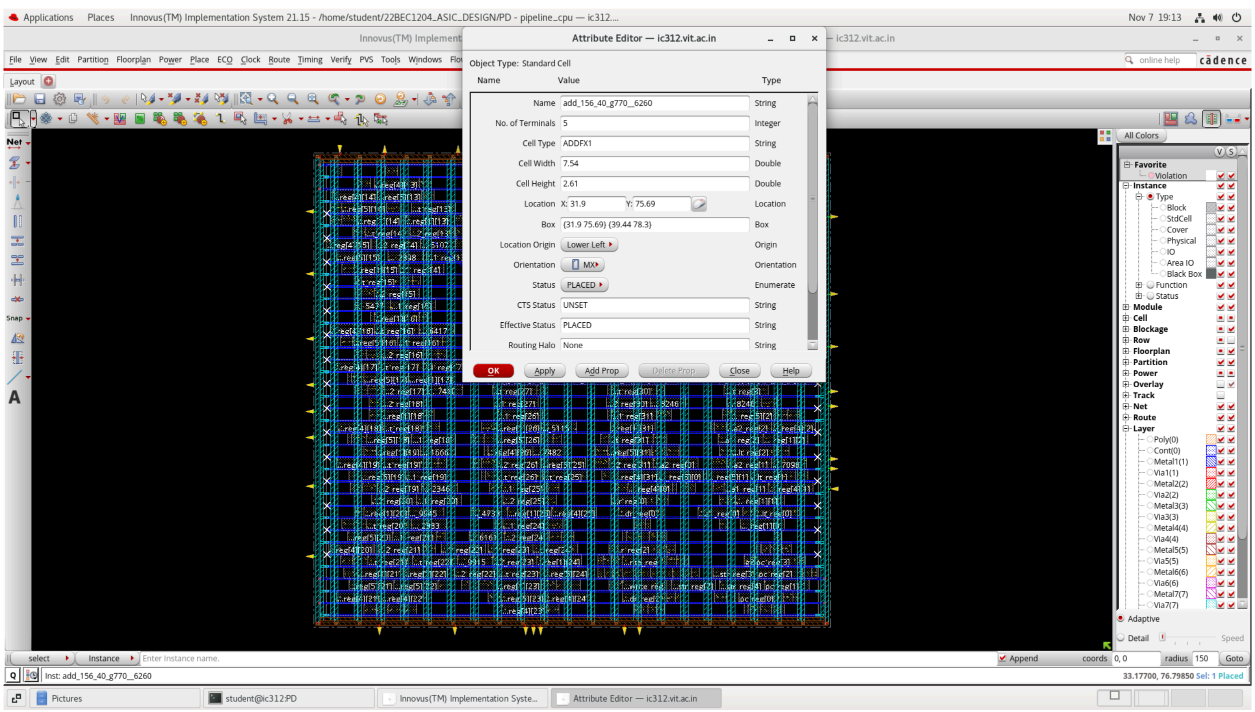

Applications
Places
Netlist Files — ic312.vit.ac.in
Nov 7 17:28
Home
22BEC1204_ASIC_DESIGN
LEC
Recent
Starred
Home
Documents
Downloads
Music
Pictures
Videos
Trash
Other Locations
Name
Size
Modified
pipeline_control_generated_LEC.log 6.7 kB Wed
pipeline_control_lec.do 512 bytes Wed
pipeline_control_generated_LEC.log~ 6.8 kB Wed
datapath_generated_LEC.log 6.0 kB Wed
datapath_lec.do 542 bytes Wed
control_unit_fsm_generated_LEC.log 5.8 kB Wed
control_unit_fsm_lec.do 550 bytes Wed
register_file_generated_LEC.log 5.7 kB Wed
register_file_lec.do 541 bytes Wed
lib_v.log 1.2 kB Wed
slowlib_verilog_generated.v 98.6 kB Wed
simple_processor_generated_LEC.log 6.0 kB 5 Sep
simple_processor_lec.do 574 bytes 5 Sep
booth_multiplier_generated_LEC.log 6.1 kB 5 Sep
booth_multiplier_generated_LEC.log~ 7.0 kB 5 Sep
Design Import — ic312.vit.ac.in
Netlist:
Verilog
LEF Files — ic312.vit.ac.in
LEF File:
NDRY/digital/90nm/dig/lef/gsclib090_macro.lef
Add
<<
LEF Files:
OUNDRY/digital/90nm/dig/lef/gsclib090_translated_ref.lef
OUNDRY/digital/90nm/dig/lef/gsclib090_translated.lef
OUNDRY/digital/90nm/dig/lef/gsclib090_tech.lef
OUNDRY/digital/90nm/dig/lef/gsclib090_macro.lef
Delete
LEF Selection:
/home/install/FOUNDRY/digital/90nm/dig/lef
doc
gsclib090_macro.lef
gsclib090_tech.lef
gsclib090_translated.lef
gsclib090_translated_ref.lef
Filters:
LEF Files (*.lef)
Close
Create Analysis Configuration ...
OK
Save...
Load...
Cancel
Help
LEC
student@ic312:PD
Innovus(TM) Implementation Syste...
Design Import — ic312.vit.ac.in

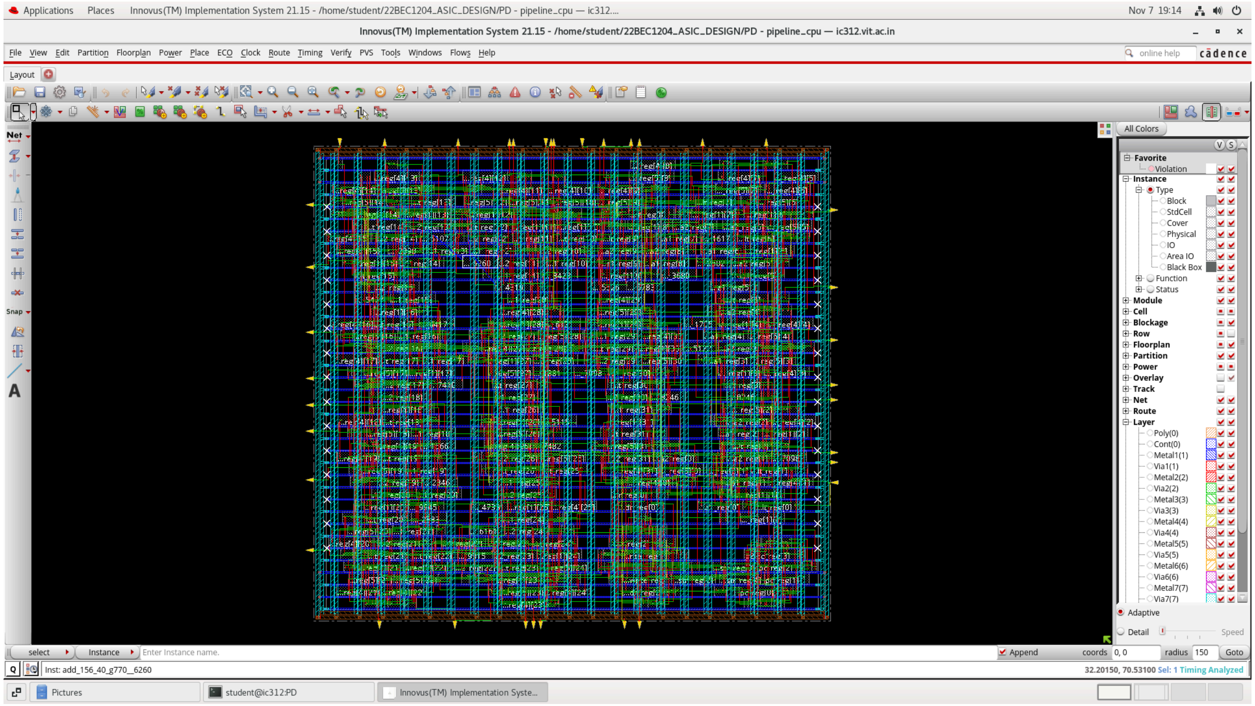

Applications
Places
MMMC Browser — ic312.vit.ac.in
Nov 7 17:31
Add RC Corner — ic312.vit.ac.in
Name: rccorner
Cap Table ../../../install/FOUNDRY/digital/90nm/dig/captable/gpdk090.lef.extended.CapTbl
Temperature:
PreRoute Resistance Scale Factor: 1.0
PreRoute Cap Scale Factor: 1.0
PreRoute Clock Resistance Scale Factor: 0.0
PreRoute Clock Cap Scale Factor: 0.0
PostRoute Resistance Scale Factor: 1.0
PostRoute Cap Scale Factor: 1.0
PostRoute Xcap Scale Factor: 1.0
PostRoute Clock Resistance Scale Factor: 0.0
PostRoute Clock Cap Scale Factor: 0.0
QRC Technology File
OK
Apply
Close
Help
LEC
student@ic312:PD
Innovus(TM) Implementation Syste...
Design Import — ic312.vit.ac.in
MMMC Browser — ic312.vit.ac.in
Add RC Corner — ic312.vit.ac.in


Add Stripes — ic312.vit.ac.in
Set Configuration
Net(s): vdd vss
Layer: Metal8(8)
Directions: Vertical Horizontal
Width: 0.7
Spacing: 0.2
Update
Set Pattern
Set-to-set distance: 5
Number of sets: 1
Bumps
Over P/G pins
Over Physical Pins
Stripe Boundary
Core ring
Pad ring:
All domains
Design boundary
Create pins
Each selected block/domain/fence
Specify rectangular area
Specify rectilinear area
First/Last Stripe
Start from: Left Right Top Bottom
Relative from core or selected area
Start: 1
Stop: 0
Absolute
OK
Apply
Defaults
Cancel
Help

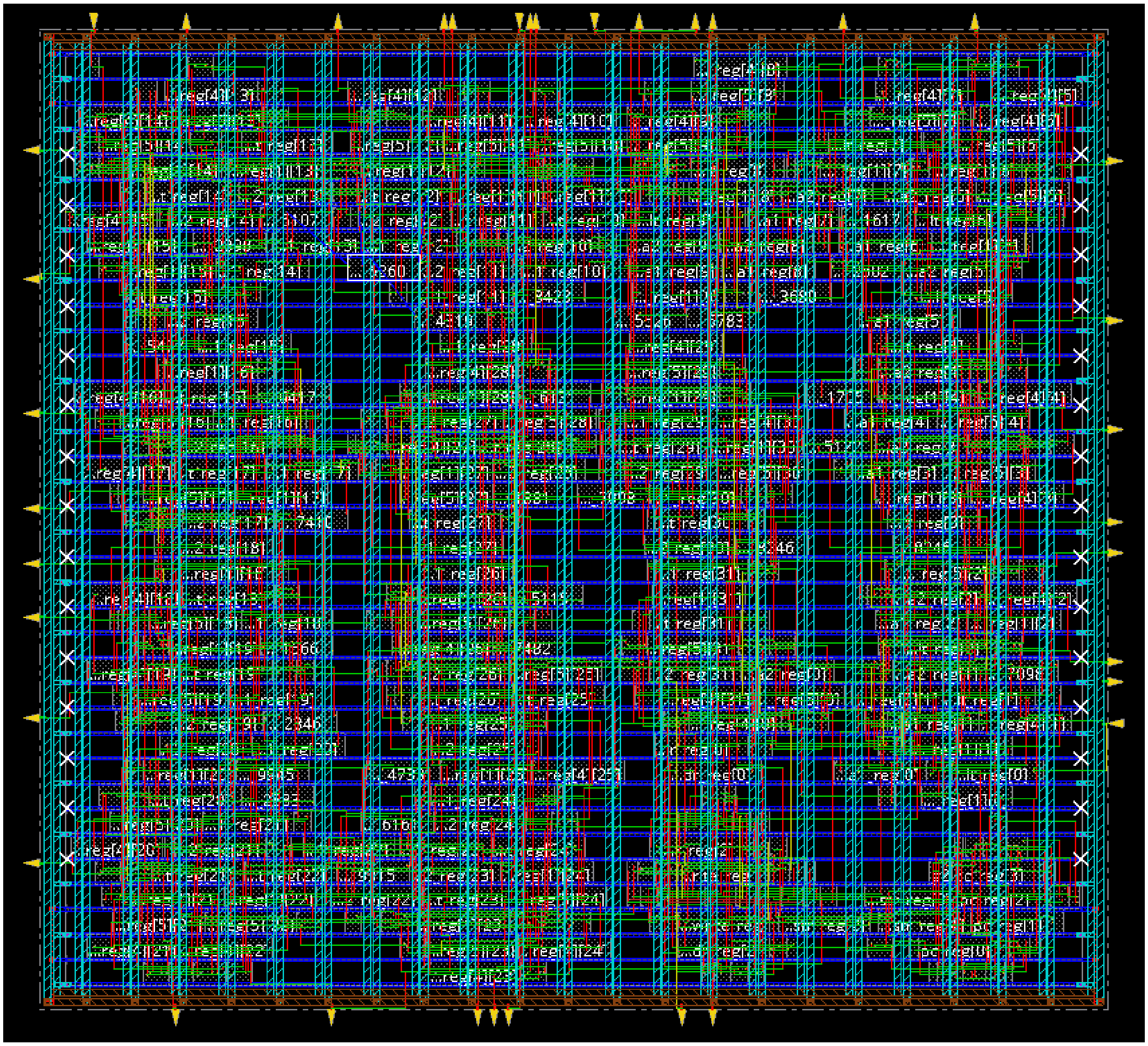

SRoute — ic312.vit.ac.in
Net(s): vdd vss
SRoute
Block Pins
Pad Rings
Floating Stripes
Pad Pins
Follow Pins
Secondary Power Pins
Routing Control
Layer Change Control
Top Layer: Metal9(9)
Bottom Layer: Metal1(1)
Allow Jogging
Allow Layer Change
Specify Area
Connect to Target Inside The Area Only
Power Domain Selection
All
Selected
Named:
Delete Existing Routes
Mode Setup
Target Editing Options
OK
Apply
Defaults
Cancel
Help

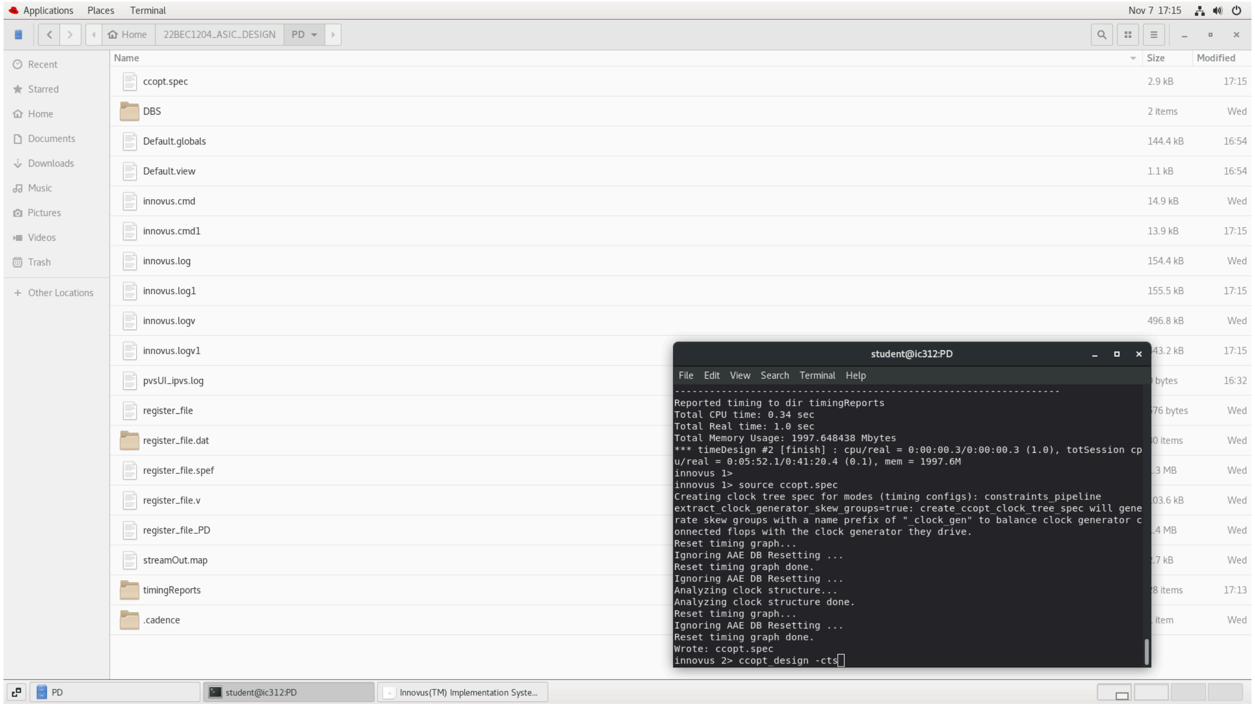

Specify Analysis Mode — ic312.vit.ac.in
Basic
Advanced
Timing Check
Setup
Hold
Timing Mode
Single
Best and Worst Case
On-Chip Variation
CPPR
Clock Gating Check
OK
Apply
Cancel
Help
Click to select single object. Shift+Click to de/select multiple objects.


Attribute Editor — ic312.vit.ac.in
Object Type: Standard Cell
Name
Value
Type
Name add_156_40_g770__6260 String
No. of Terminals 5 Integer
Cell Type ADDFX1 String
Cell Width 7.54 Double
Cell Height 2.61 Double
Location X: 31.9 Y: 75.69 Location
Box {31.9 75.69} {39.44 78.3} Box
Location Origin Lower Left Origin
Orientation MX Orientation
Status PLACED Enumerate
CTS Status UNSET String
Effective Status PLACED String
Routing Halo None String
OK
Apply
Add Prop
Delete Prop
Close
Help
Inst: add_156_40_g770__6260



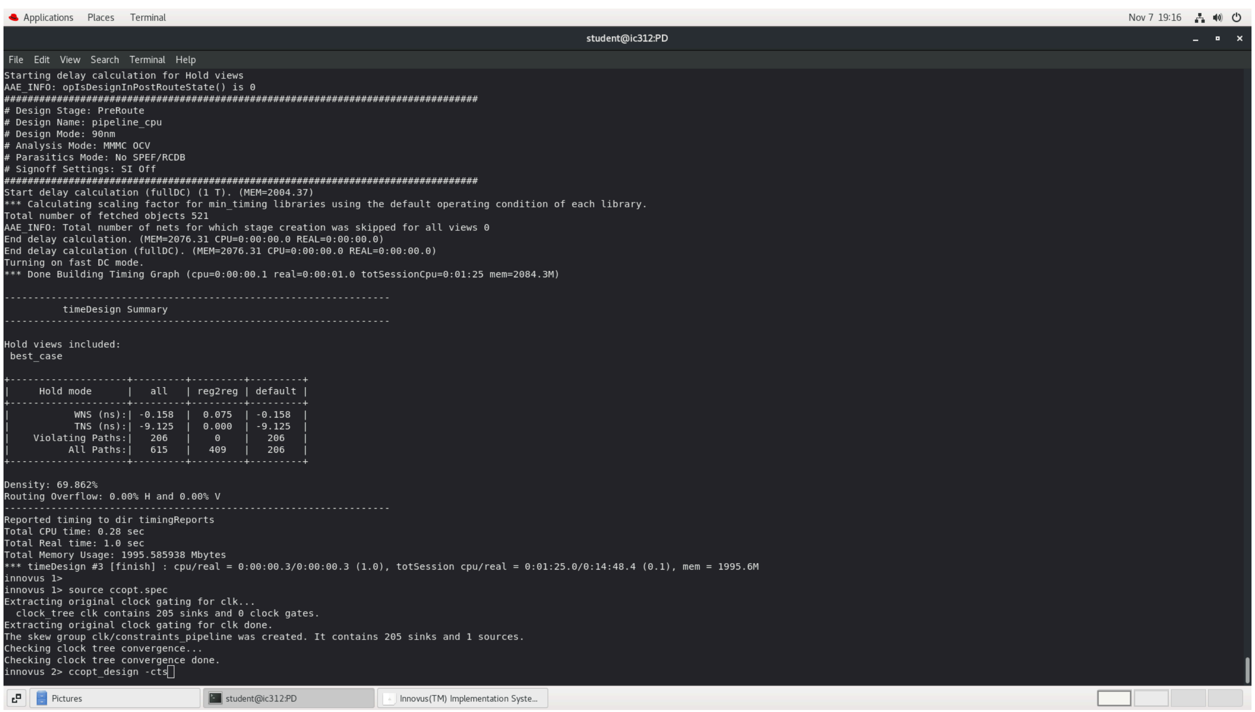
Innovus(TM) Implementation System 21.15 - /home/student/22BEC1204_ASIC_DESIGN/PD - pipeline_cpu — ic312.vit.ac.in





Applications Places Terminal
Nov 7 17:15
Home 22BEC1204_ASIC_DESIGN PD
Recent
Starred
Home
Documents
Downloads
Music
Pictures
Videos
Trash
Other Locations
Name Size Modified
ccopt.spec 2.9 kB 17:15
DBS 2 items Wed
Default.globals 144.4 kB 16:54
Default.view 1.1 kB 16:54
innovus.cmd 14.9 kB Wed
innovus.cmd1 13.9 kB 17:15
innovus.log 154.4 kB Wed
innovus.log1 155.5 kB 17:15
innovus.logv 496.8 kB Wed
innovus.logv1
pvsUI_ipvs.log
register_file
register_file.dat
register_file.spef
register_file.v
register_file_PD
streamOut.map
timingReports
.cadence
student@ic312:PD
File Edit View Search Terminal Help

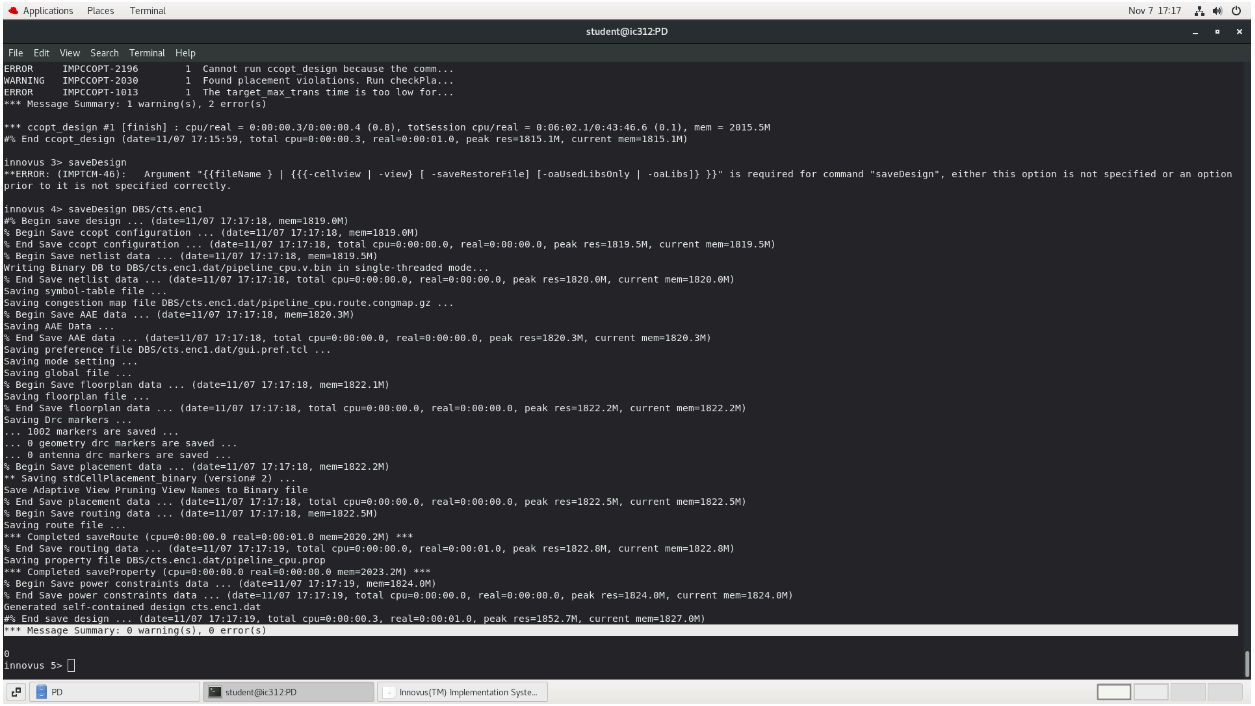
Reported timing to dir timingReports
Total CPU time: 0.34 sec
Total Real time: 1.0 sec
Total Memory Usage: 1997.648438 Mbytes
*** timeDesign #2 [finish] : cpu/real = 0:00:00.3/0:00:00.3 (1.0), totSession cp
u/real = 0:05:52.1/0:41:20.4 (0.1), mem = 1997.6M
innovus 1>
innovus 1> source ccopt.spec
Creating clock tree spec for modes (timing configs): constraints_pipeline
extract_clock_generator_skew_groups=true: create_ccopt_clock_tree_spec will gene
rate skew groups with a name prefix of "_clock_gen" to balance clock generator c
onnected flops with the clock generator they drive.
Reset timing graph...
Ignoring AAE DB Resetting ...
Reset timing graph done.
Ignoring AAE DB Resetting ...
Analyzing clock structure...
Analyzing clock structure done.
Reset timing graph...
Ignoring AAE DB Resetting ...
Reset timing graph done.
Wrote: ccopt.spec
innovus 2> ccopt_design -cts
PD
student@ic312:PD
Innovus(TM) Implementation Syste...

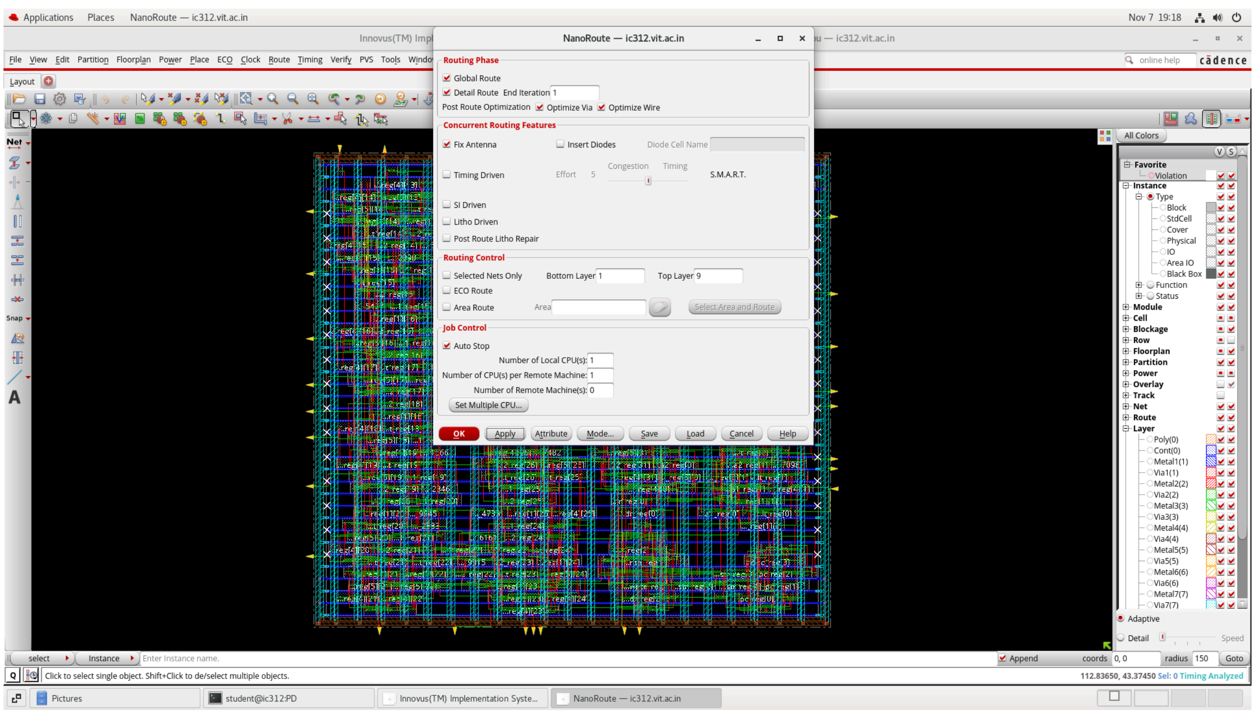


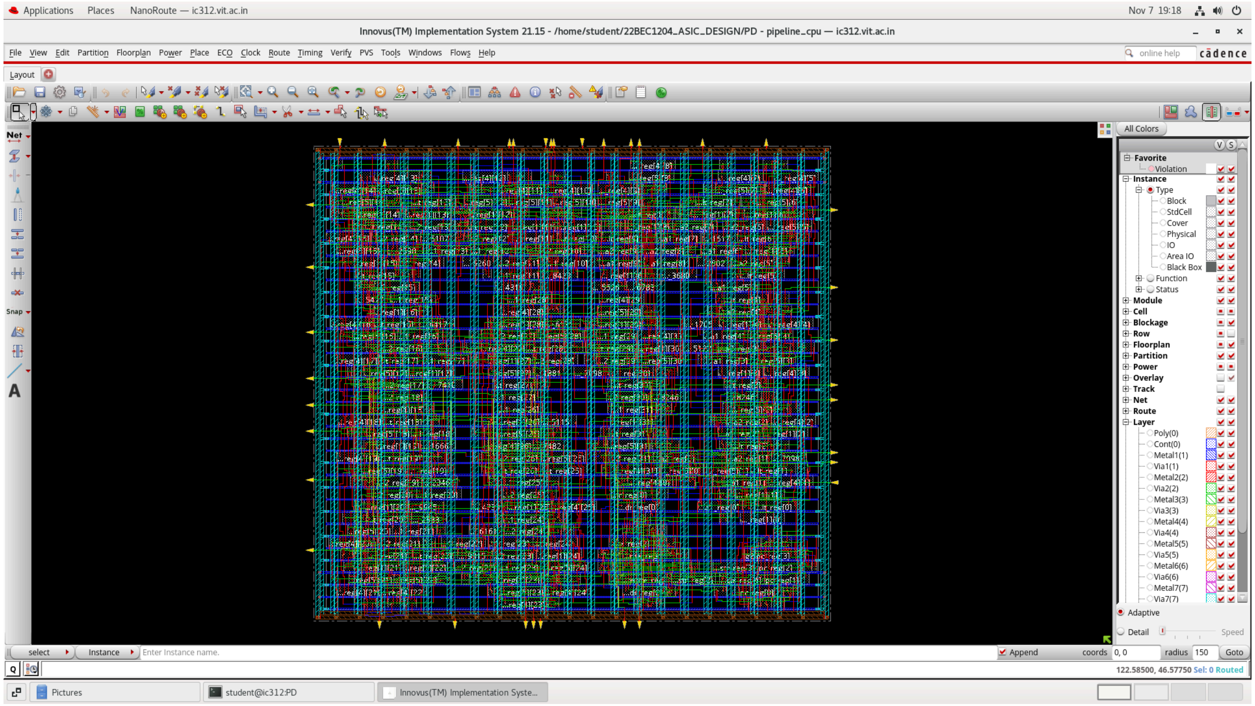
Applications Places Terminal
Nov 7 19:16
student@ic312:PD
File Edit View Search Terminal Help
Starting delay calculation for Hold views
AAE_INFO: opIsDesignInPostRouteState() is 0
##
Design Stage: PreRoute
Design Name: pipeline_cpu
Design Mode: 90nm
Analysis Mode: MMMC OCV
Parasitics Mode: No SPEF/RCDB
Signoff Settings: SI Off
##
Start delay calculation (fullDC) (1 T). (MEM=2004.37)
*** Calculating scaling factor for min_timing libraries using the default operating condition of each library.
Total number of fetched objects 521
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=2076.31 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=2076.31 CPU=0:00:00.0 REAL=0:00:00.0)
Turning on fast DC mode.
*** Done Building Timing Graph (cpu=0:00:00.1 real=0:00:01.0 totSessionCpu=0:01:25 mem=2084.3M)
----
timeDesign Summary

Hold views included:
best_case
+--------------------+---------+---------+---------+
| Hold mode | all | reg2reg | default |
+--------------------+---------+---------+---------+
WNS (ns):	-0.158	0.075	-0.158
TNS (ns):	-9.125	0.000	-9.125
Violating Paths:	206	0	206
All Paths:	615	409	206
+--------------------+---------+---------+---------+
Density: 69.862%
Routing Overflow: 0.00% H and 0.00% V

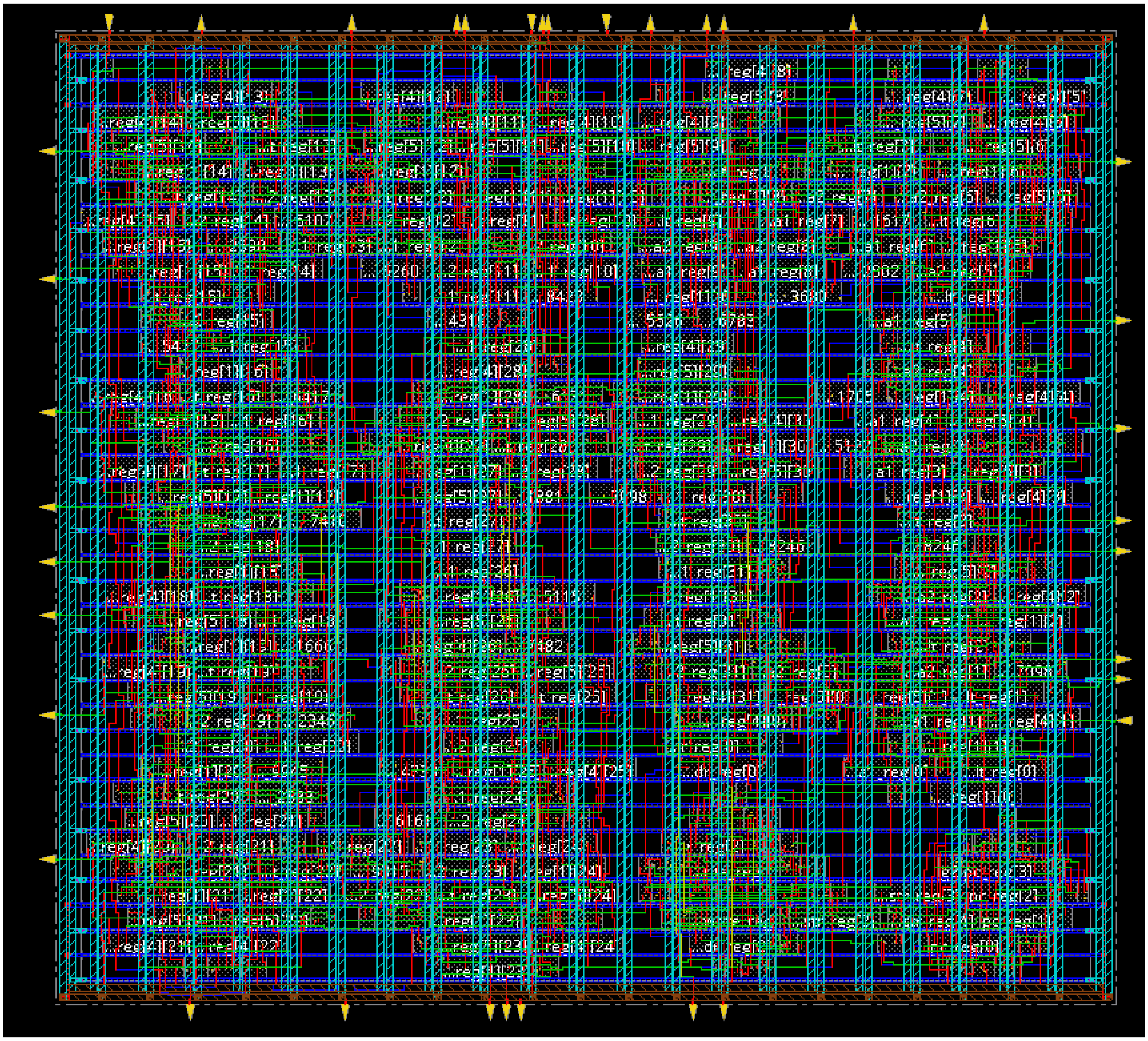
Reported timing to dir timingReports
Total CPU time: 0.28 sec
Total Real time: 1.0 sec
Total Memory Usage: 1995.585938 Mbytes
*** timeDesign #3 [finish] : cpu/real = 0:00:00.3/0:00:00.3 (1.0), totSession cpu/real = 0:01:25.0/0:14:48.4 (0.1), mem = 1995.6M
innovus 1>
innovus 1> source ccopt.spec
Extracting original clock gating for clk...
clock_tree clk contains 205 sinks and 0 clock gates.
Extracting original clock gating for clk done.
The skew group clk/constraints_pipeline was created. It contains 205 sinks and 1 sources.
Checking clock tree convergence...
Checking clock tree convergence done.
innovus 2> ccopt_design -cts
Pictures
student@ic312:PD
Innovus(TM) Implementation Syste...

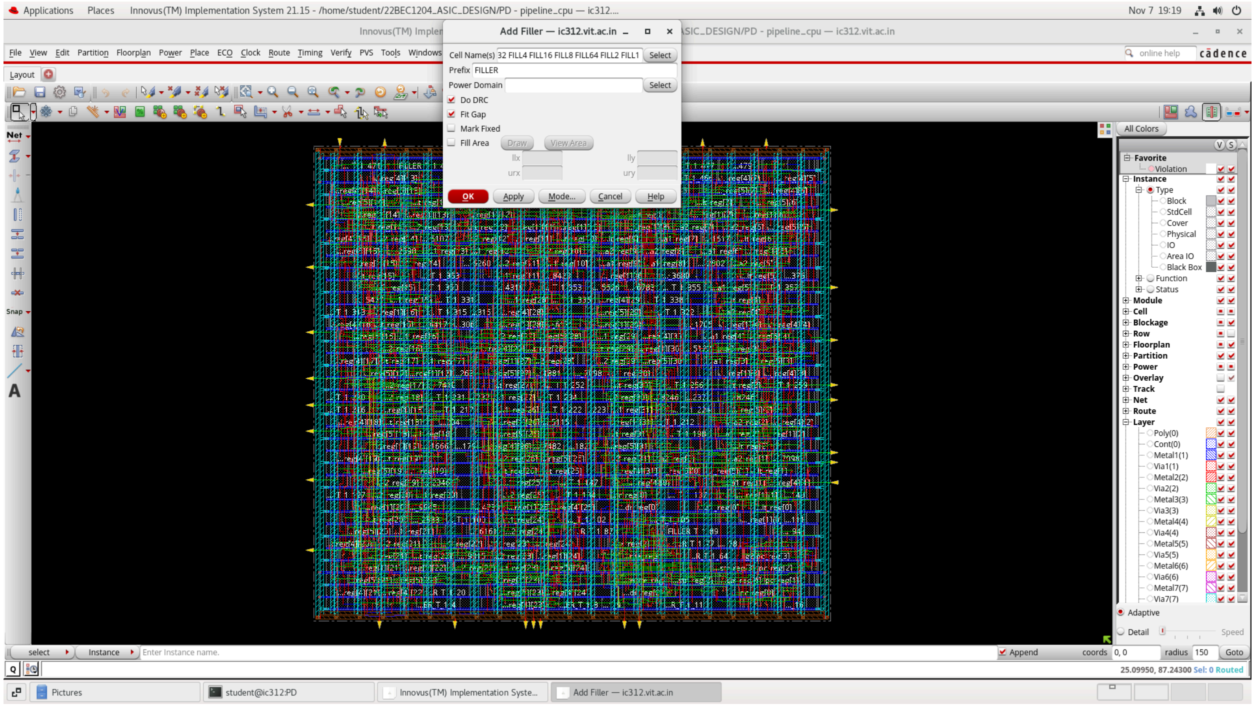

Applications Places Terminal
Nov 7 17:17
student@ic312:PD
File Edit View Search Terminal Help
ERROR IMPCCOPT-2196 1 Cannot run ccopt_design because the comm...
WARNING IMPCCOPT-2030 1 Found placement violations. Run checkPla...
ERROR IMPCCOPT-1013 1 The target_max_trans time is too low for...
*** Message Summary: 1 warning(s), 2 error(s)
*** ccopt_design #1 [finish] : cpu/real = 0:00:00.3/0:00:00.4 (0.8), totSession cpu/real = 0:06:02.1/0:43:46.6 (0.1), mem = 2015.5M
#% End ccopt_design (date=11/07 17:15:59, total cpu=0:00:00.3, real=0:00:01.0, peak res=1815.1M, current mem=1815.1M)
innovus 3> saveDesign
**ERROR: (IMPTCM-46): Argument "{{fileName } | {{{-cellview | -view} [-saveRestoreFile] [-oaUsedLibsOnly | -oaLibs]} }}" is required for command "saveDesign", either this option is not specified or an option prior to it is not specified correctly.
innovus 4> saveDesign DBS/cts.enc1
#% Begin save design ... (date=11/07 17:17:18, mem=1819.0M)
% Begin Save ccopt configuration ... (date=11/07 17:17:18, mem=1819.0M)
% End Save ccopt configuration ... (date=11/07 17:17:18, total cpu=0:00:00.0, real=0:00:00.0, peak res=1819.5M, current mem=1819.5M)
% Begin Save netlist data ... (date=11/07 17:17:18, mem=1819.5M)
Writing Binary DB to DBS/cts.enc1.dat/pipeline_cpu.v.bin in single-threaded mode...
% End Save netlist data ... (date=11/07 17:17:18, total cpu=0:00:00.0, real=0:00:00.0, peak res=1820.0M, current mem=1820.0M)
Saving symbol-table file ...
Saving congestion map file DBS/cts.enc1.dat/pipeline_cpu.route.congmap.gz ...
% Begin Save AAE data ... (date=11/07 17:17:18, mem=1820.3M)
Saving AAE Data ...
% End Save AAE data ... (date=11/07 17:17:18, total cpu=0:00:00.0, real=0:00:00.0, peak res=1820.3M, current mem=1820.3M)
Saving preference file DBS/cts.enc1.dat/gui.pref.tcl ...
Saving mode setting ...
Saving global file ...
% Begin Save floorplan data ... (date=11/07 17:17:18, mem=1822.1M)
Saving floorplan file ...
% End Save floorplan data ... (date=11/07 17:17:18, total cpu=0:00:00.0, real=0:00:00.0, peak res=1822.2M, current mem=1822.2M)
Saving Drc markers ...
... 1002 markers are saved ...
... 0 geometry drc markers are saved ...
... 0 antenna drc markers are saved ...
% Begin Save placement data ... (date=11/07 17:17:18, mem=1822.2M)
** Saving stdCellPlacement_binary (version# 2) ...
Save Adaptive View Pruning View Names to Binary file
% End Save placement data ... (date=11/07 17:17:18, total cpu=0:00:00.0, real=0:00:00.0, peak res=1822.5M, current mem=1822.5M)
% Begin Save routing data ... (date=11/07 17:17:18, mem=1822.5M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:01.0 mem=2020.2M) ***
% End Save routing data ... (date=11/07 17:17:19, total cpu=0:00:00.0, real=0:00:01.0, peak res=1822.8M, current mem=1822.8M)
Saving property file DBS/cts.enc1.dat/pipeline_cpu.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=2023.2M) ***
% Begin Save power constraints data ... (date=11/07 17:17:19, mem=1824.0M)
% End Save power constraints data ... (date=11/07 17:17:19, total cpu=0:00:00.0, real=0:00:00.0, peak res=1824.0M, current mem=1824.0M)
Generated self-contained design cts.enc1.dat
#% End save design ... (date=11/07 17:17:19, total cpu=0:00:00.3, real=0:00:01.0, peak res=1852.7M, current mem=1827.0M)
*** Message Summary: 0 warning(s), 0 error(s)
0
innovus 5>
PD
student@ic312:PD
Innovus(TM) Implementation Syste...

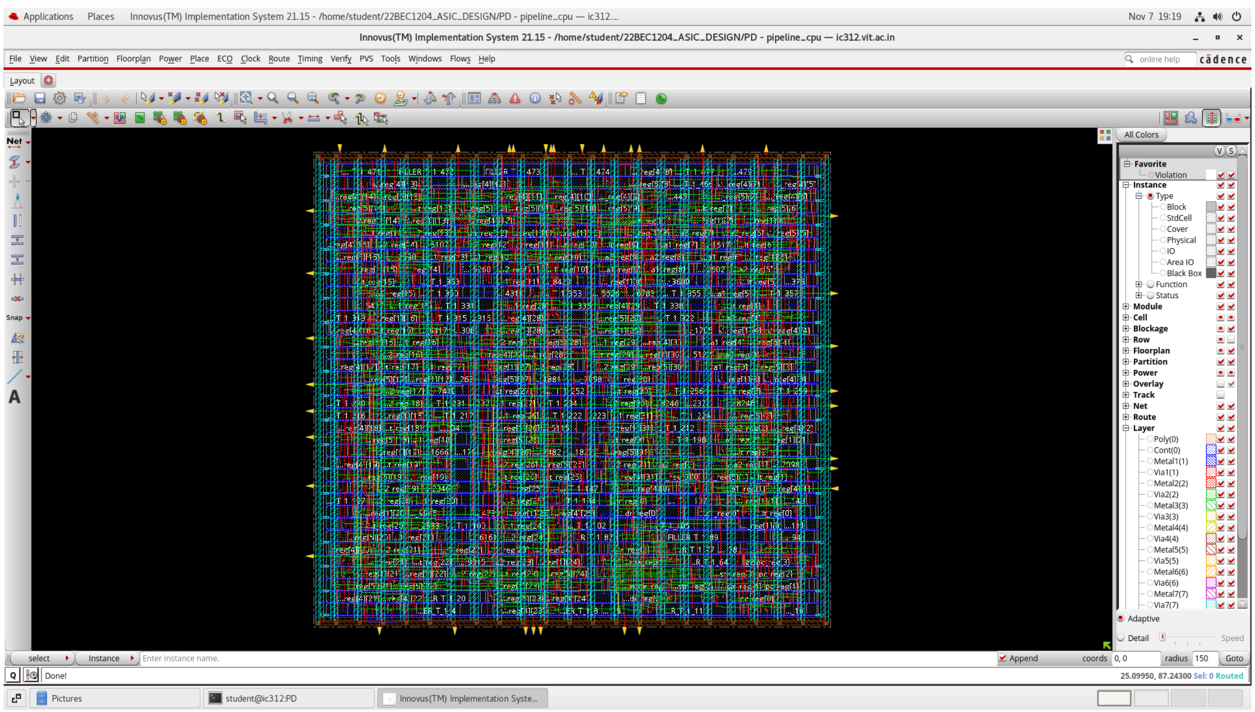

Applications Places NanoRoute — ic312.vit.ac.in
Nov 7 19:18
NanoRoute — ic312.vit.ac.in
Routing Phase
Global Route
Detail Route End Iteration 1
Post Route Optimization Optimize Via Optimize Wire
Concurrent Routing Features
Fix Antenna
Insert Diodes
Diode Cell Name
Timing Driven
Effort 5
Congestion
Timing
S.M.A.R.T.
SI Driven
Litho Driven
Post Route Litho Repair
Routing Control
Selected Nets Only
Bottom Layer 1
Top Layer 9
ECO Route
Area Route
Area
Select Area and Route
Job Control
Auto Stop
Number of Local CPU(s): 1
Number of CPU(s) per Remote Machine: 1
Number of Remote Machine(s): 0
Set Multiple CPU...
OK
Apply
Attribute
Mode...
Save
Load
Cancel
Help
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools
cadence
Layout
All Colors
Favorite
Violation
Instance
Type
Block
StdCell
Cover
Physical
IO
Area IO
Black Box
Function
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Net
Route
Layer
Poly(0)
Cont(0)
Metal1(1)
Via1(1)
Metal2(2)
Via2(2)
Metal3(3)
Via3(3)
Metal4(4)
Via4(4)
Metal5(5)
Via5(5)
Metal6(6)
Via6(6)
Metal7(7)
Via7(7)
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords 0, 0
radius 150
Goto
Click to select single object. Shift+Click to de/select multiple objects.
112.83650, 43.37450 Sel: 0 Timing Analyzed
Pictures
student@ic312:PD
Innovus(TM) Implementation Syste...
NanoRoute — ic312.vit.ac.in

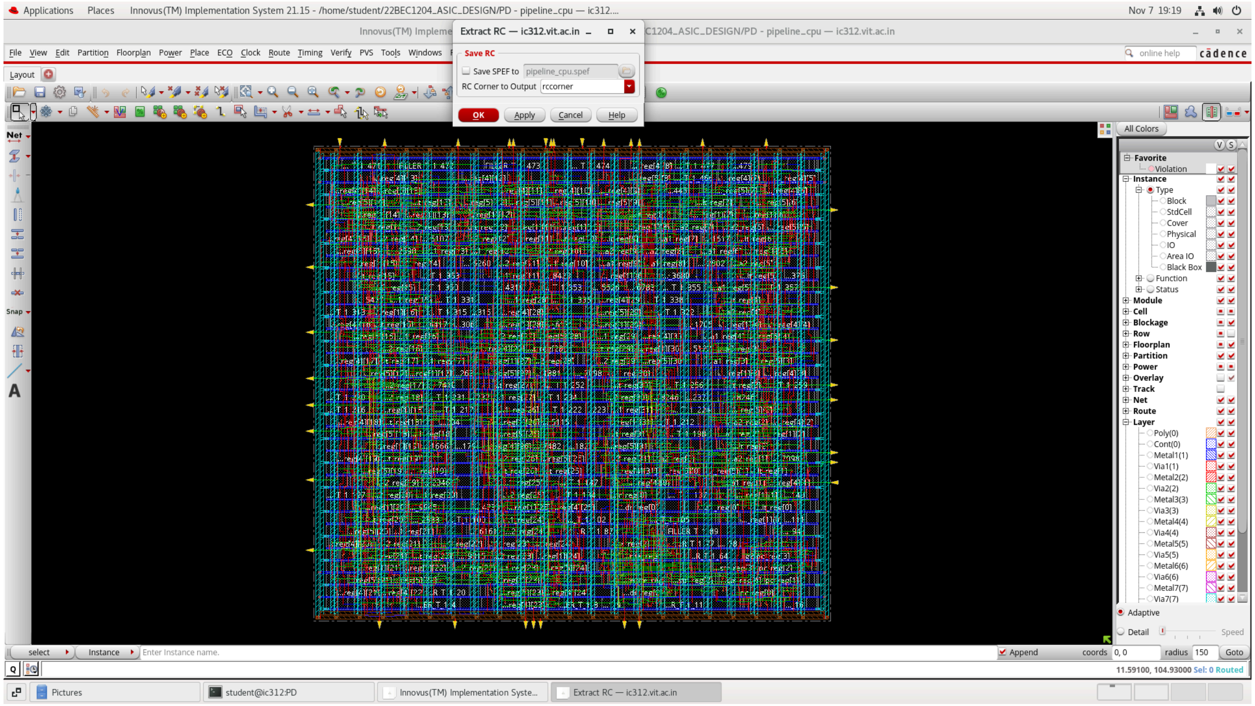


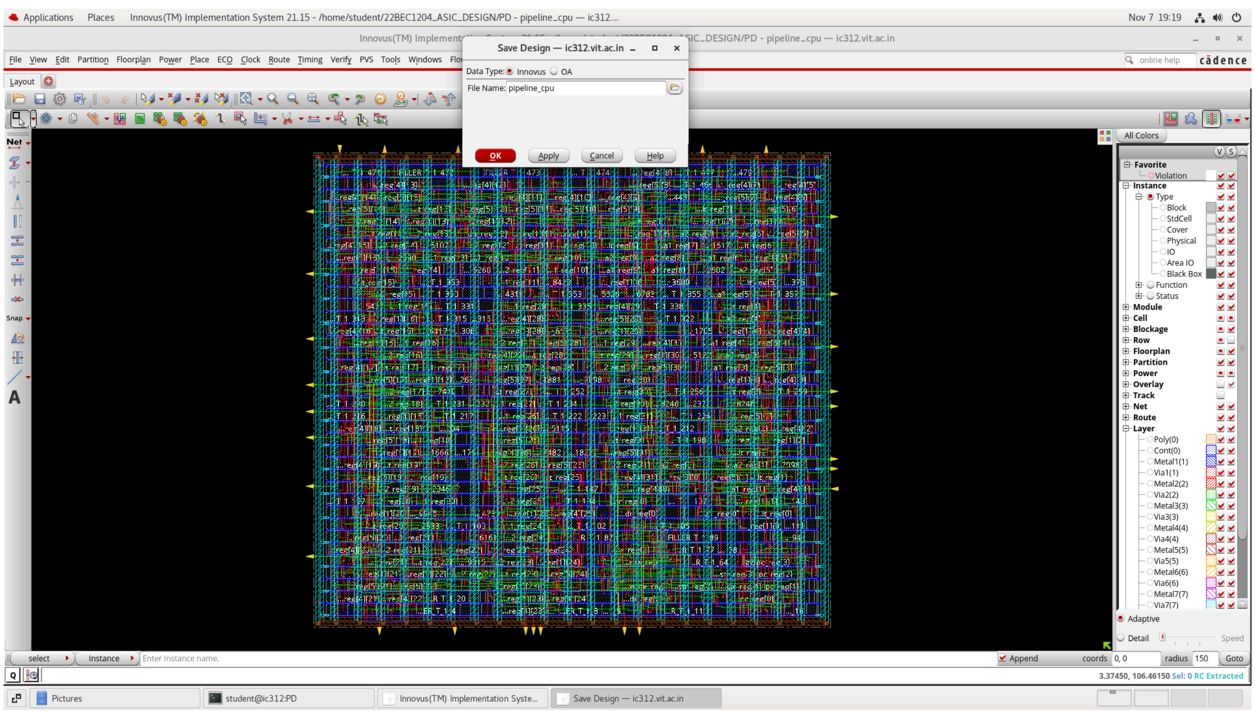
Applications
Places
NanoRoute — ic312.vit.ac.in
Nov 7 19:18
Innovus(TM) Implementation System 21.15 - /home/student/22BEC1204_ASIC_DESIGN/PD - pipeline_cpu — ic312.vit.ac.in
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flows Help
online help
cādence
Layout
All Colors
Favorite
Violation
Instance
Type
Block
StdCell
Cover
Physical
IO
Area IO
Black Box
Function
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Net
Route
Layer
Poly(0)
Cont(0)
Metal1(1)
Via1(1)
Metal2(2)
Via2(2)
Metal3(3)
Via3(3)
Metal4(4)
Via4(4)
Metal5(5)
Via5(5)
Metal6(6)
Via6(6)
Metal7(7)
Via7(7)
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords
0, 0
radius
150
Goto
122.58500, 46.57750 Sel: 0 Routed
Pictures
student@ic312:PD
Innovus(TM) Implementation Syste...

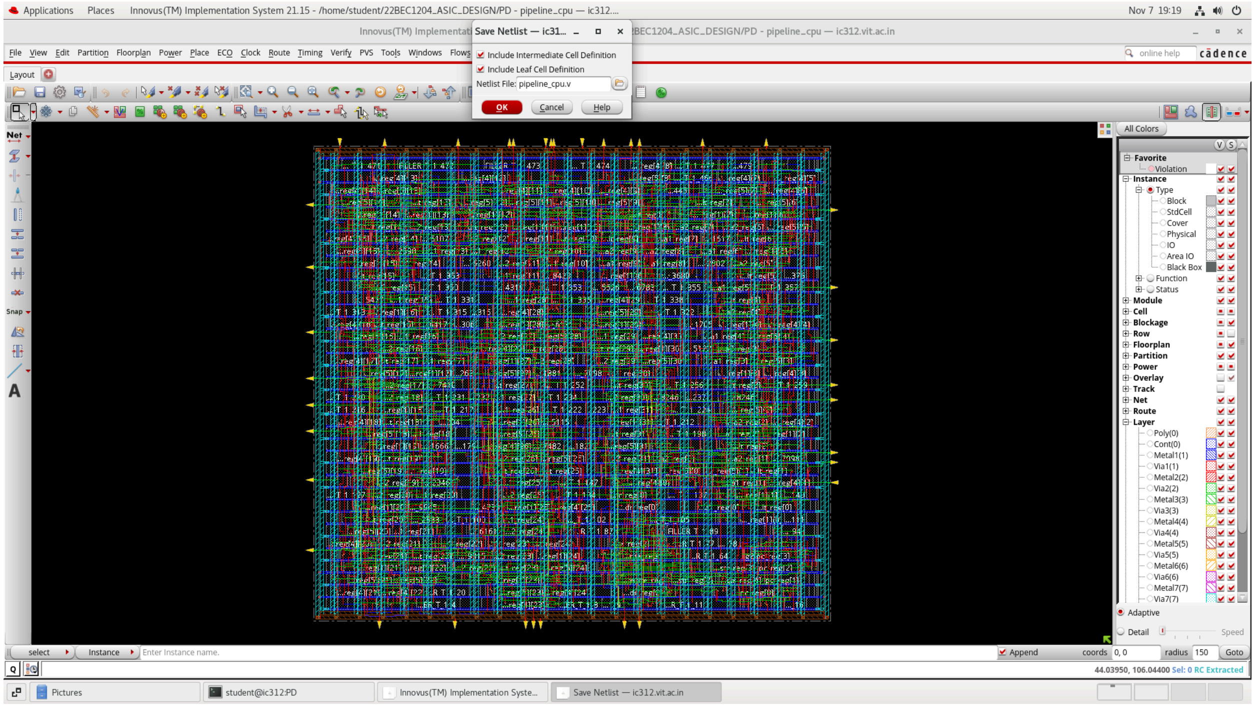


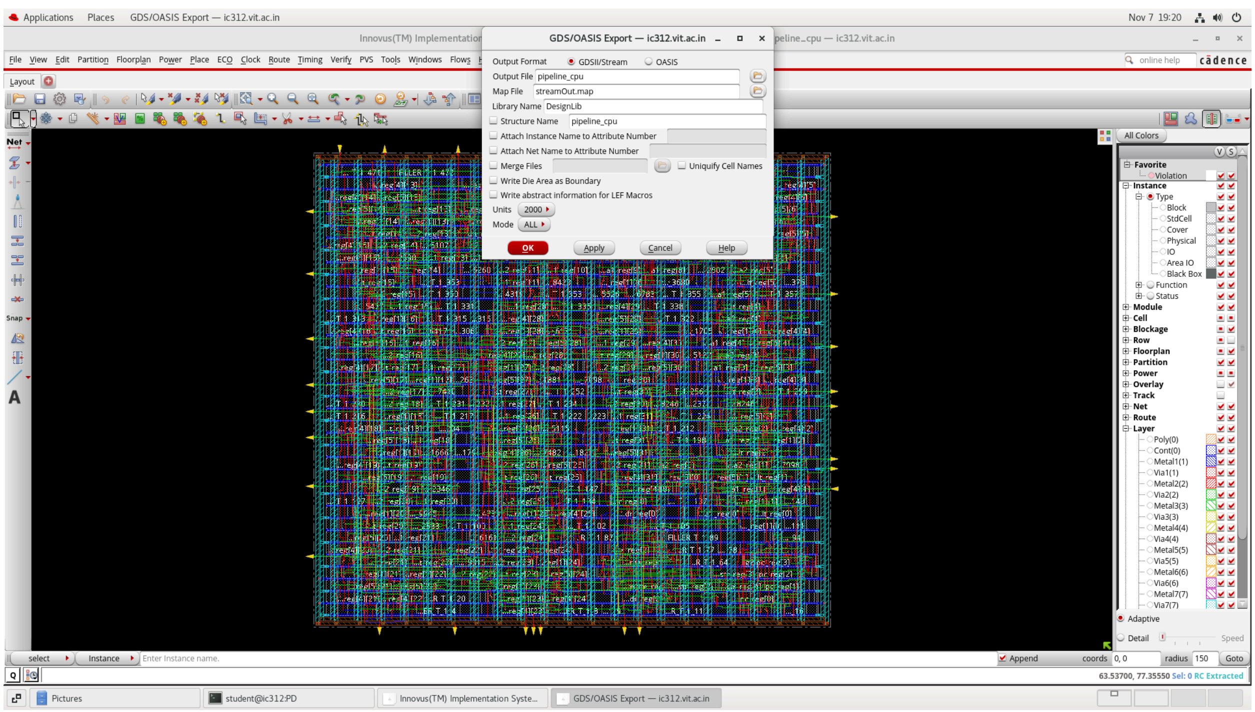

Applications Places Innovus(TM) Implementation System 21.15 - /home/student/22BEC1204_ASIC_DESIGN/PD - pipeline_cpu — ic312...
Nov 7 19:19
Add Filler — ic312.vit.ac.in
Cell Name(s) 32 FILL4 FILL16 FILL8 FILL64 FILL2 FILL1
Prefix FILLER
Power Domain
Do DRC
Fit Gap
Mark Fixed
Fill Area
OK
Apply
Mode...
Cancel
Help
All Colors
Favorite
Instance
Layer
25.09950, 87.24300 Sel: 0 Routed

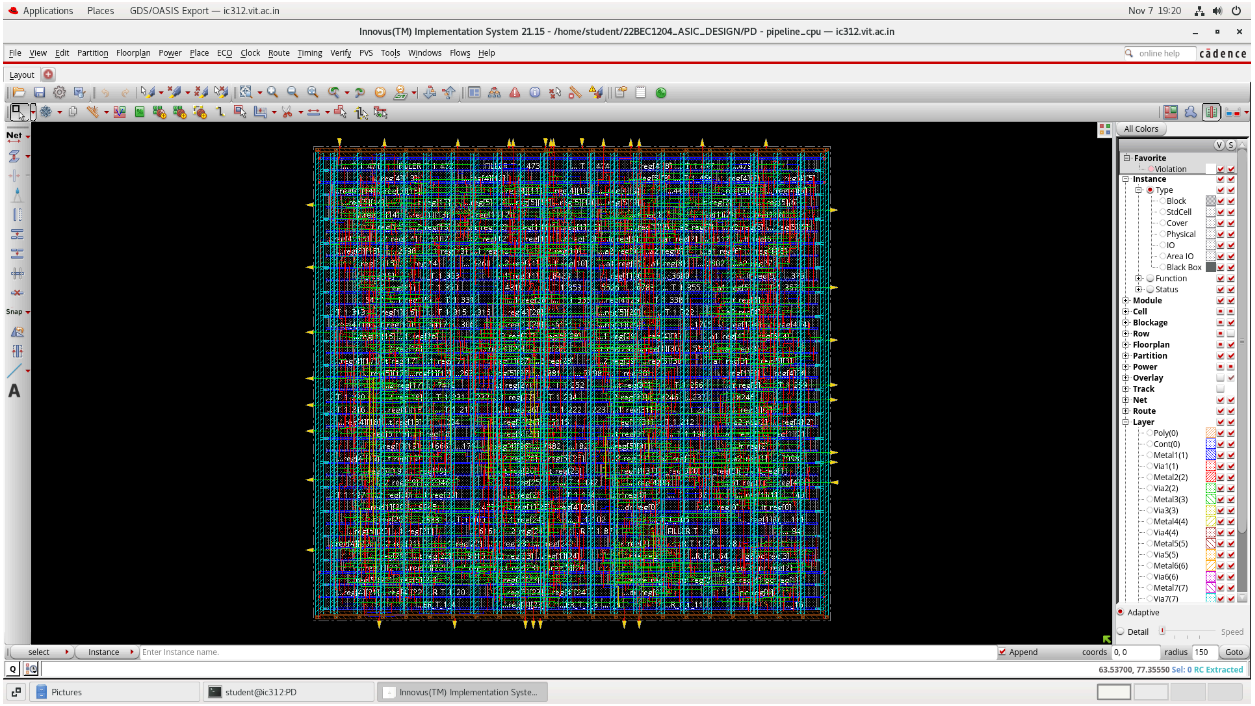

Applications Places Innovus(TM) Implementation System 21.15 - /home/student/22BEC1204_ASIC_DESIGN/PD - pipeline_cpu — ic312...
Nov 7 19:19
Innovus(TM) Implementation System 21.15 - /home/student/22BEC1204_ASIC_DESIGN/PD - pipeline_cpu — ic312.vit.ac.in
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flows Help
Done!
25.09950, 87.24300 Sel: 0 Routed


Innovus(TM) Implementation System 21.15 - /home/student/22BEC1204_ASIC_DESIGN/PD - pipeline_cpu — ic312...
Nov 7 19:19
Extract RC — ic312.vit.ac.in
Save RC
Save SPEF to
pipeline_cpu.spef
RC Corner to Output
rccorner
OK
Apply
Cancel
Help
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows
Layout
cadence
All Colors
Favorite
Violation
Instance
Type
Block
StdCell
Cover
Physical
IO
Area IO
Black Box
Function
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Net
Route
Layer
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords
radius
150
Goto
11.59100, 104.93000 Sel: 0 Routed
Pictures
student@ic312:PD
Innovus(TM) Implementation Syste...
Extract RC — ic312.vit.ac.in

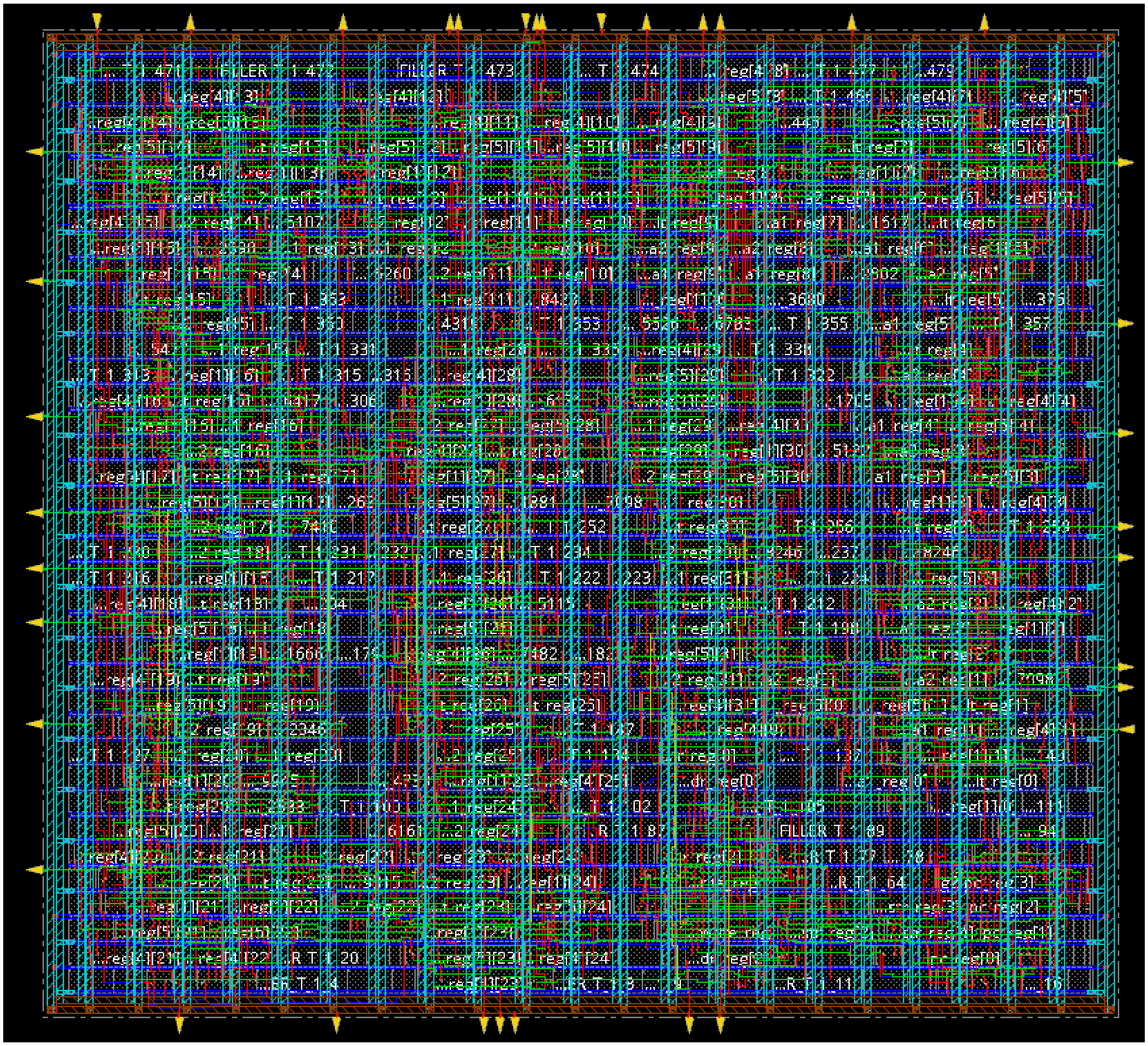

Innovus(TM) Implementation System 21.15 - /home/student/22BEC1204_ASIC_DESIGN/PD - pipeline_cpu — ic312...
Nov 7 19:19
Save Design — ic312.vit.ac.in
Data Type:
Innovus
OA
File Name: pipeline_cpu
OK
Apply
Cancel
Help
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows
Layout
cadence
All Colors
select
Instance
Enter Instance name.
Append
coords
radius
150
Goto
3.37450, 106.46150 Sel: 0 RC Extracted
Pictures
student@ic312:PD
Innovus(TM) Implementation Syste...
Save Design — ic312.vit.ac.in







Applications
Places
GDS/OASIS Export — ic312.vit.ac.in
Nov 7 19:20
Innovus(TM) Implementation System 21.15 - /home/student/22BEC1204_ASIC_DESIGN/PD - pipeline_cpu — ic312.vit.ac.in
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flows Help
online help
cadence
Layout
All Colors
Favorite
Violation
Instance
Type
Block
StdCell
Cover
Physical
IO
Area IO
Black Box
Function
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Net
Route
Layer
Poly(0)
Cont(0)
Metal1(1)
Via1(1)
Metal2(2)
Via2(2)
Metal3(3)
Via3(3)
Metal4(4)
Via4(4)
Metal5(5)
Via5(5)
Metal6(6)
Via6(6)
Metal7(7)
Via7(7)
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords
0, 0
radius
150
Goto
63.53700, 77.35550 Sel: 0 RC Extracted
Pictures
student@ic312:PD
Innovus(TM) Implementation Syste...



7. Inference:

A 4-stage pipelined CPU controller with hazard detection was successfully designed, verified, and implemented using a complete ASIC flow. The project highlighted critical differences between simulation and synthesis:

1. **Synthesizable Logic:** initial blocks were replaced with a synthesizable case statement for the ROM.
2. **Logic Preservation:** An output port was required to prevent synthesis tools from optimizing away the design.
3. **LEC Consistency:** Reset strategies must be consistent across all sequential elements (all asynchronous or all synchronous) to ensure a successful logic equivalence check.

The design passed all stages, from functional simulation to a final, routed GDSII file, demonstrating a successful RTL-to-GDSII implementation.